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(54) **AIRCRAFT RUDDER WITH ARTICULATED
SUPPORT STRUCTURE OF PEDAL**

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B64C 13/04 (2006.01)

B64C 13/06 (2006.01)

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(52) **U.S. Cl.**

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(2013.01); **B64C 13/12** (2013.01); **B64C 25/44**
(2013.01)

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B64C 25/44

See application file for complete search history.

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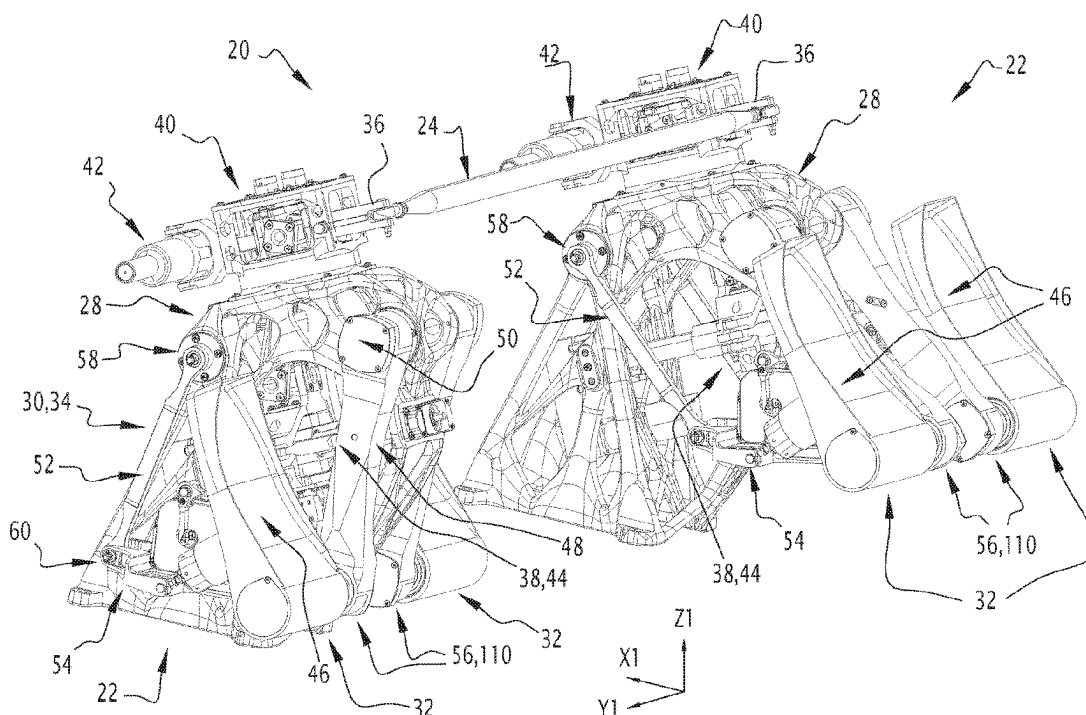
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(57)

ABSTRACT

An aircraft rudder pedal comprising a frame and at least one pedal side system. The or each pedal side system includes a pedal and an articulated support structure of the pedal connecting the pedal to the frame. The articulated support structure of said pedal comprises a crank, a lever and a support link. The crank presents a first pivotal articulation connection to the frame and a second pivotal articulation connection with the support link. The lever presents a first pivotal articulation connection with the frame and a second pivotal articulation connection with the support link.

20 Claims, 10 Drawing Sheets



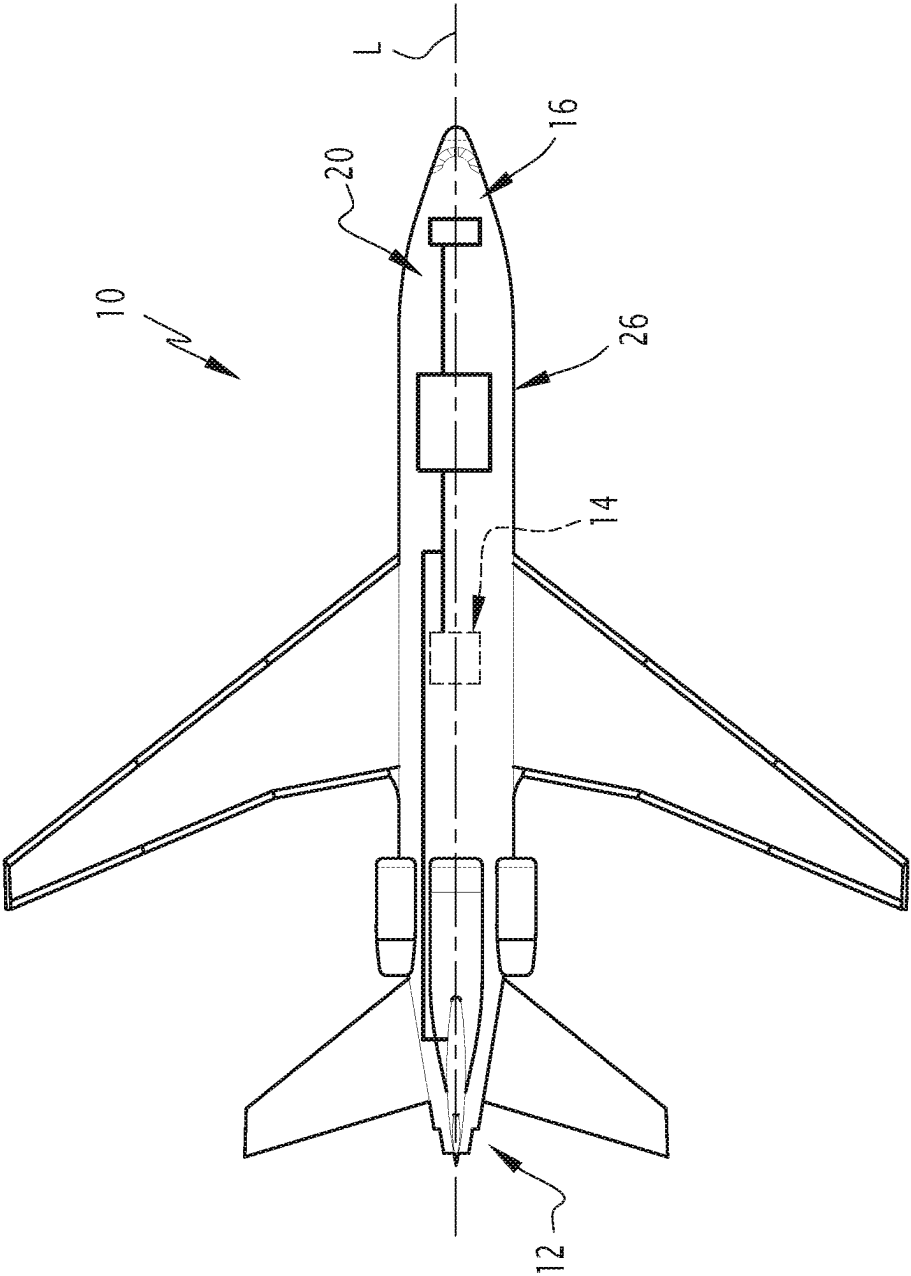


FIG.1

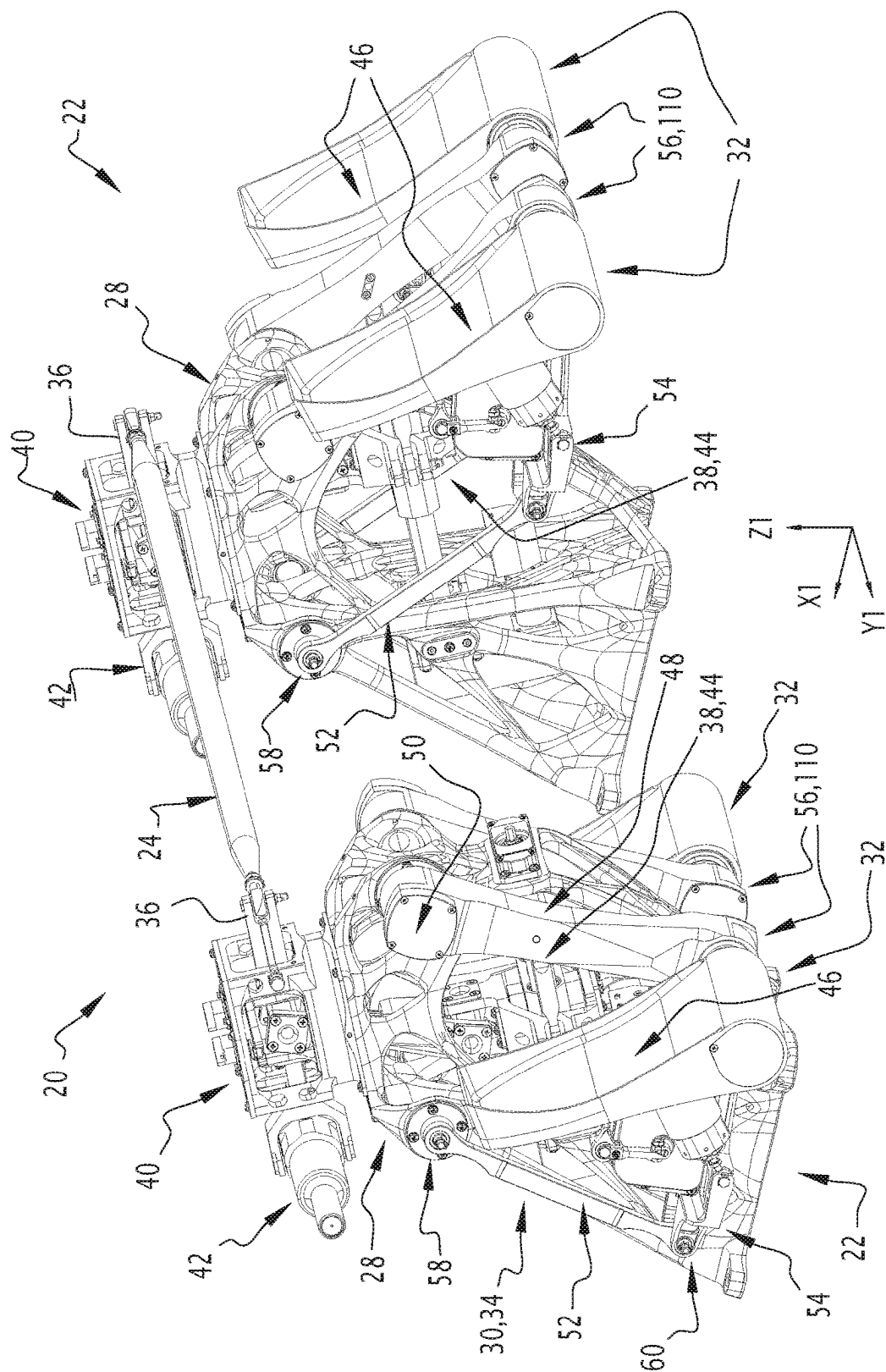


FIG. 2

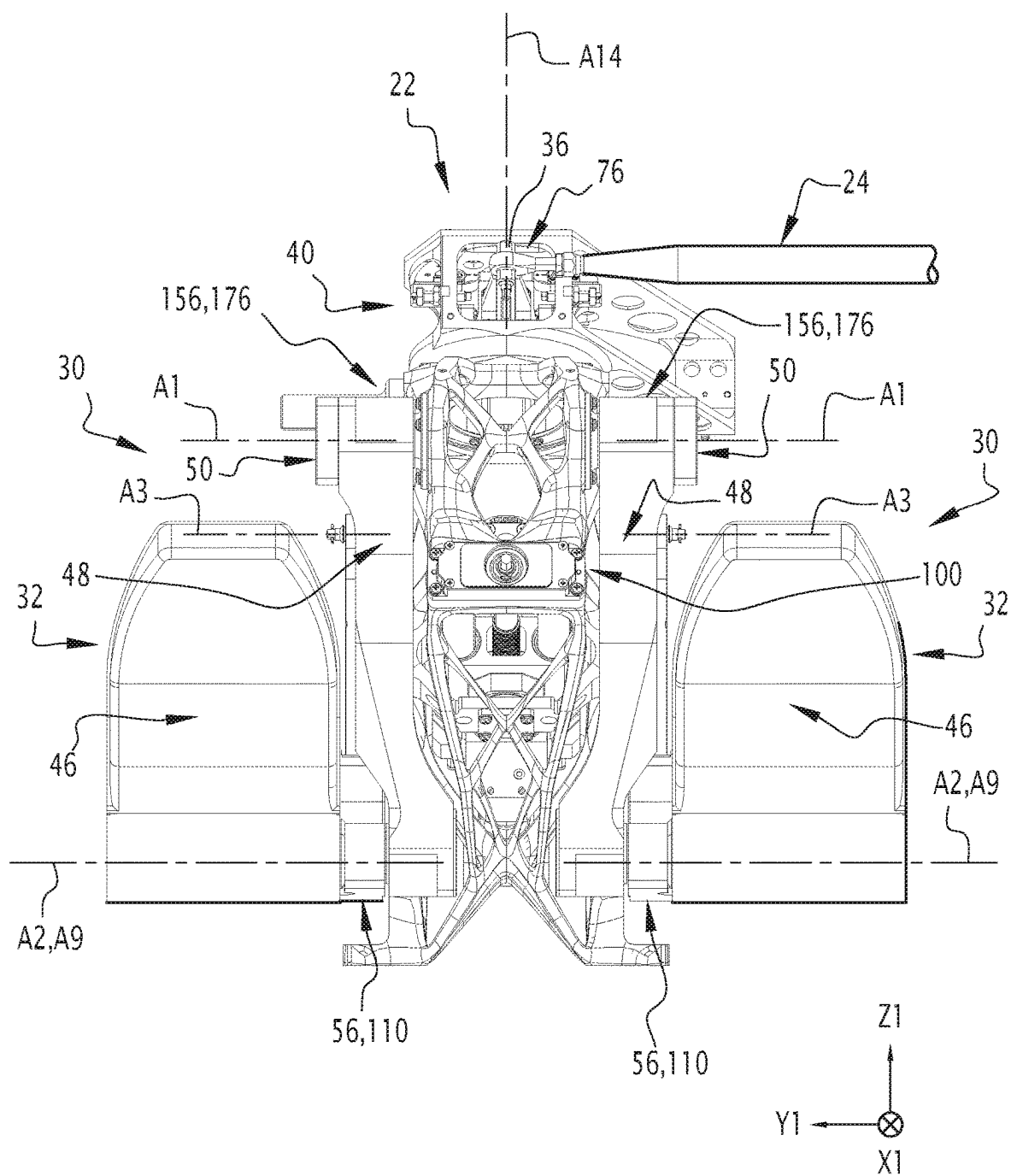


FIG. 3

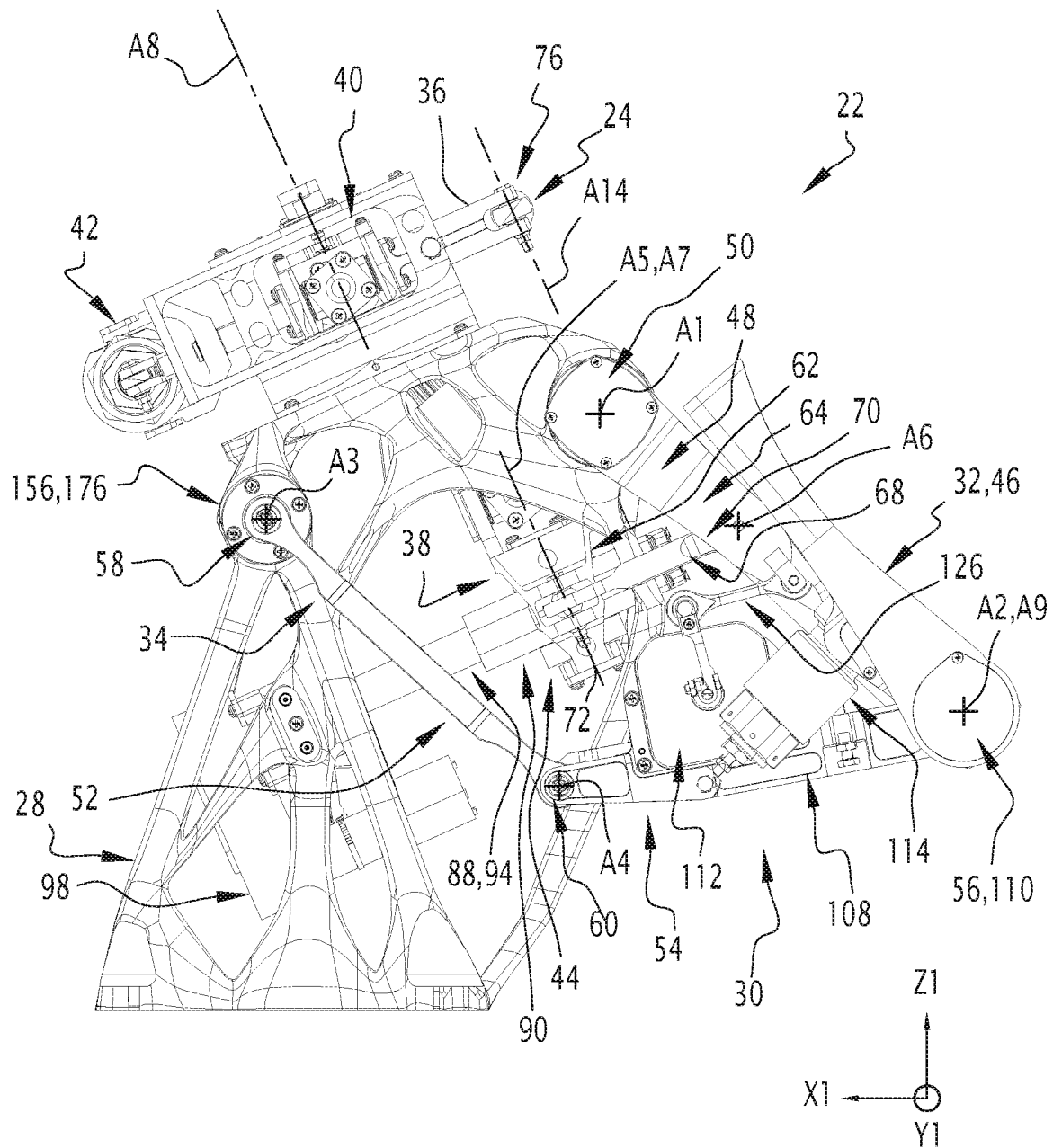


FIG.4

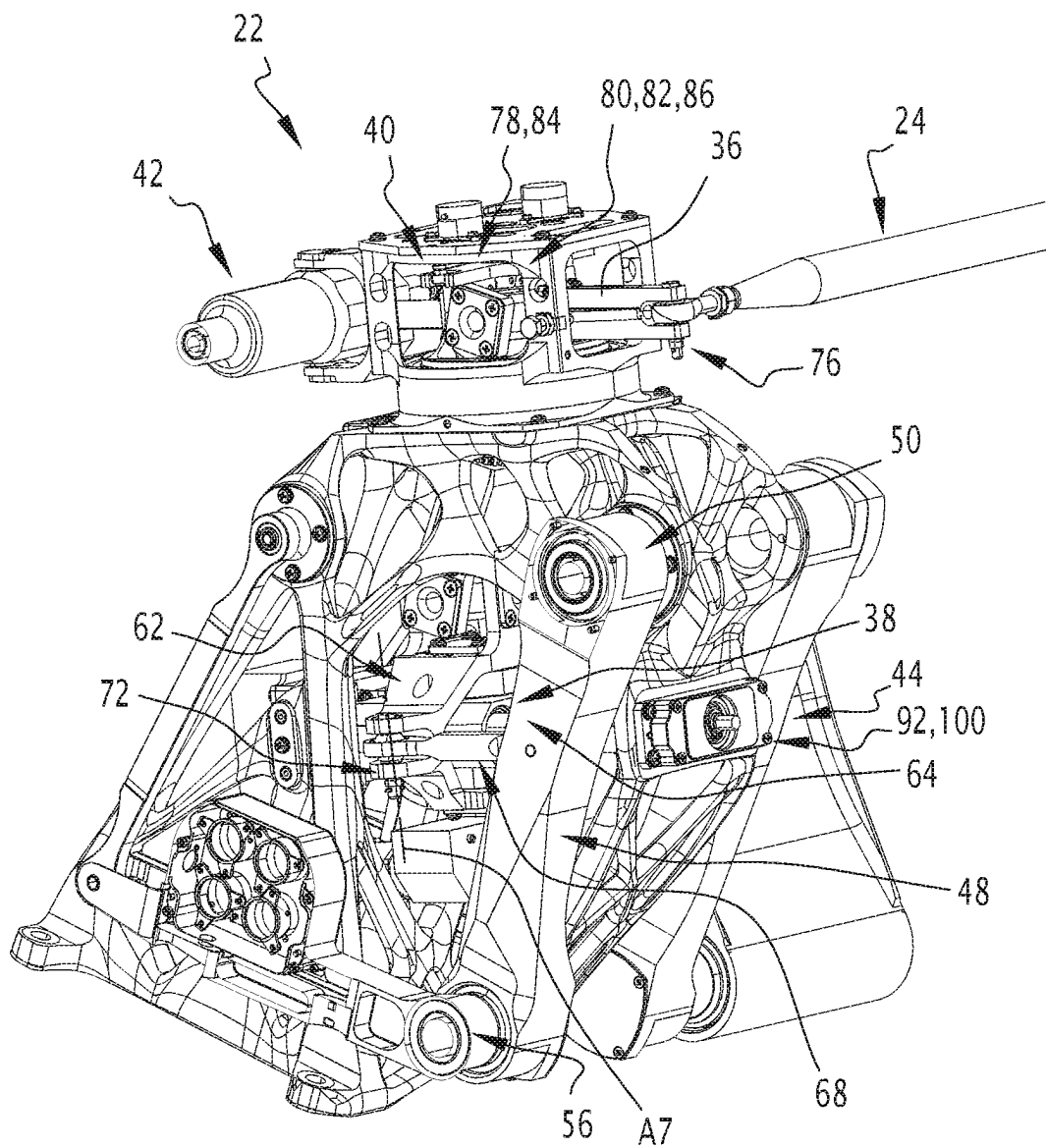


FIG. 5

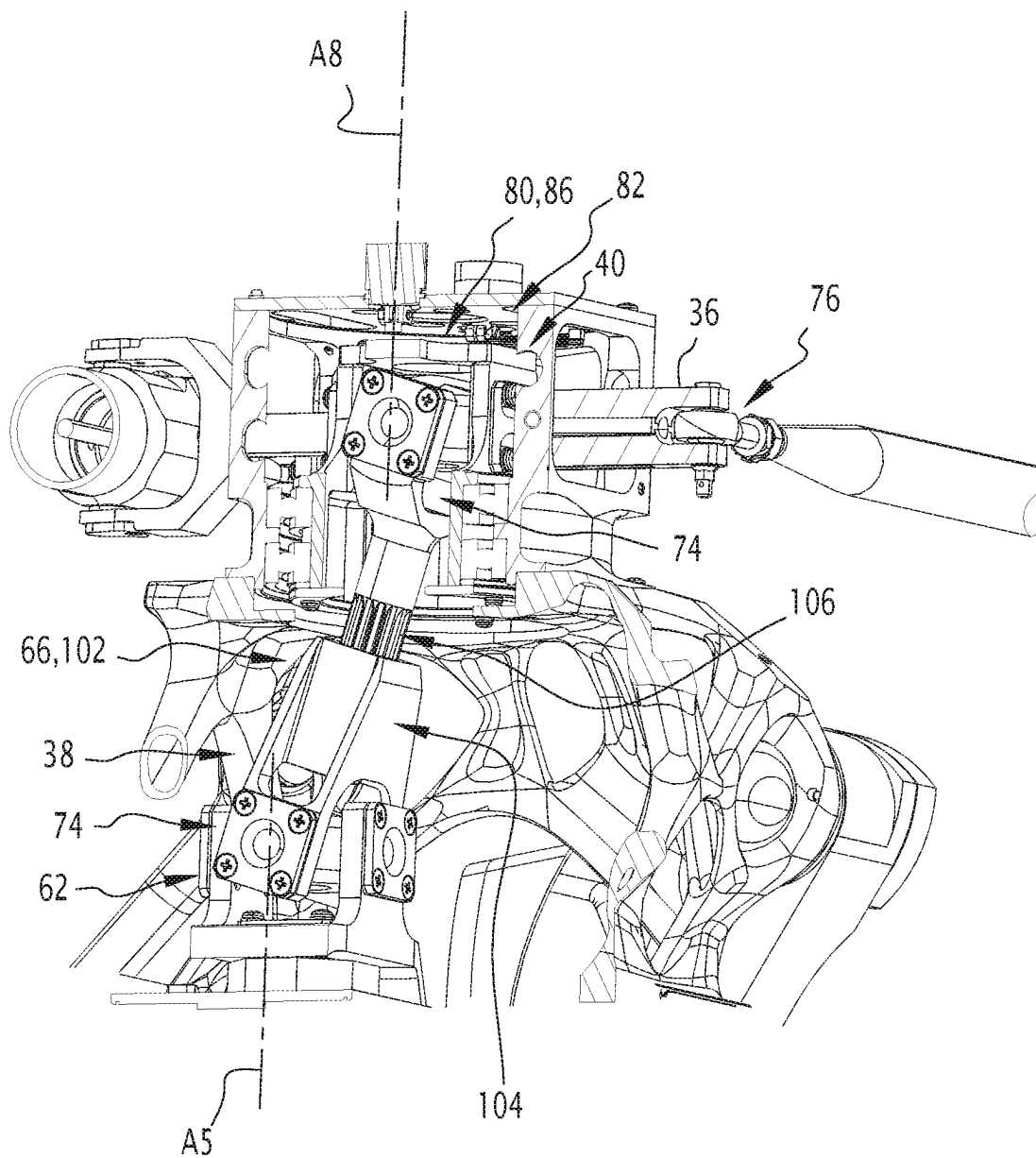


FIG. 6

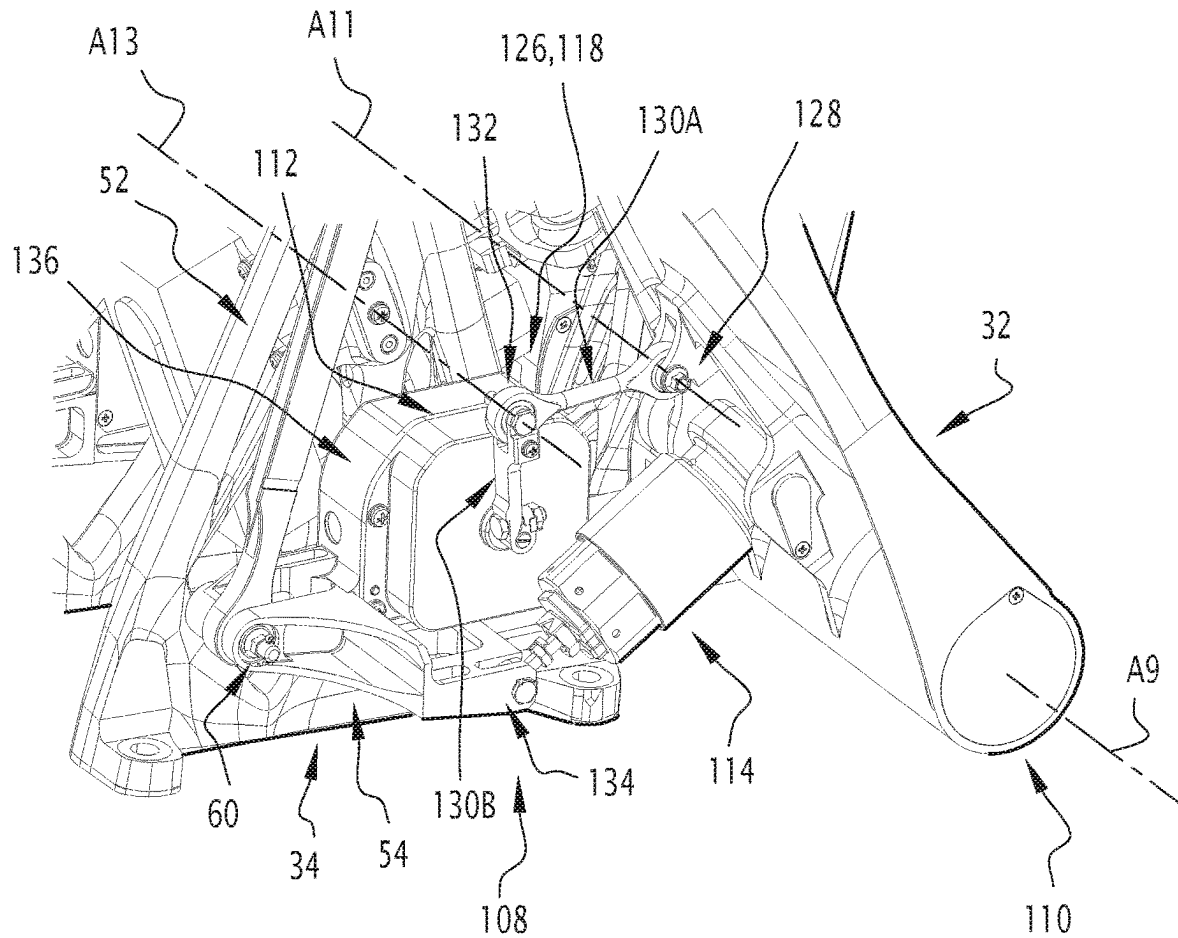


FIG. 7

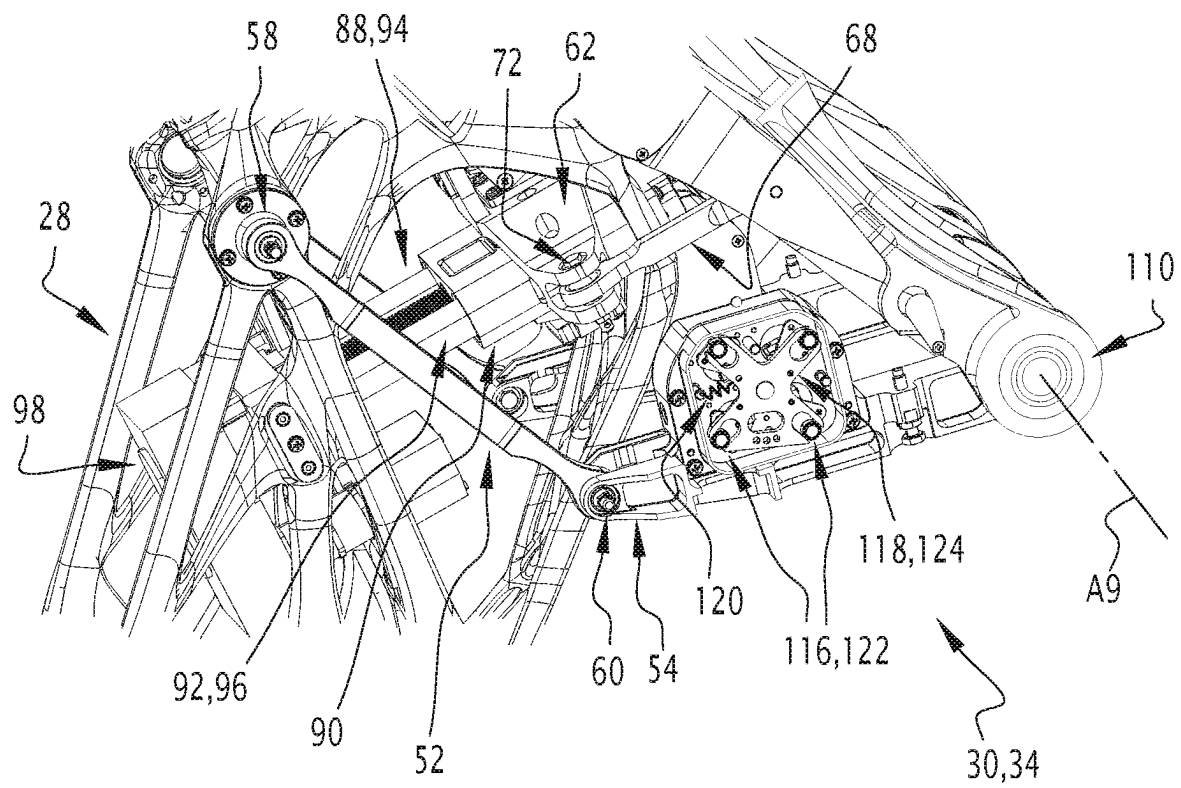


FIG. 8

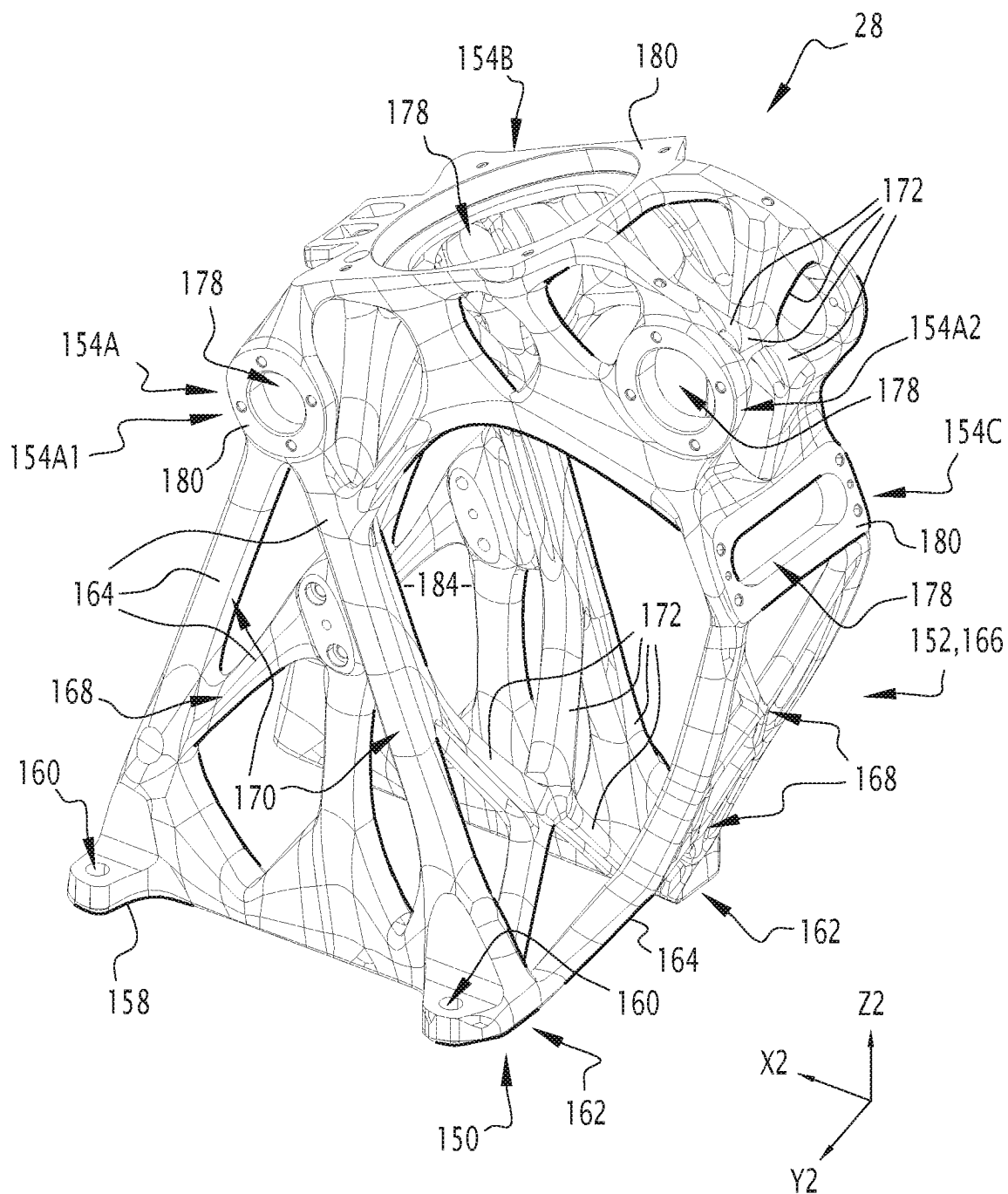


FIG. 9

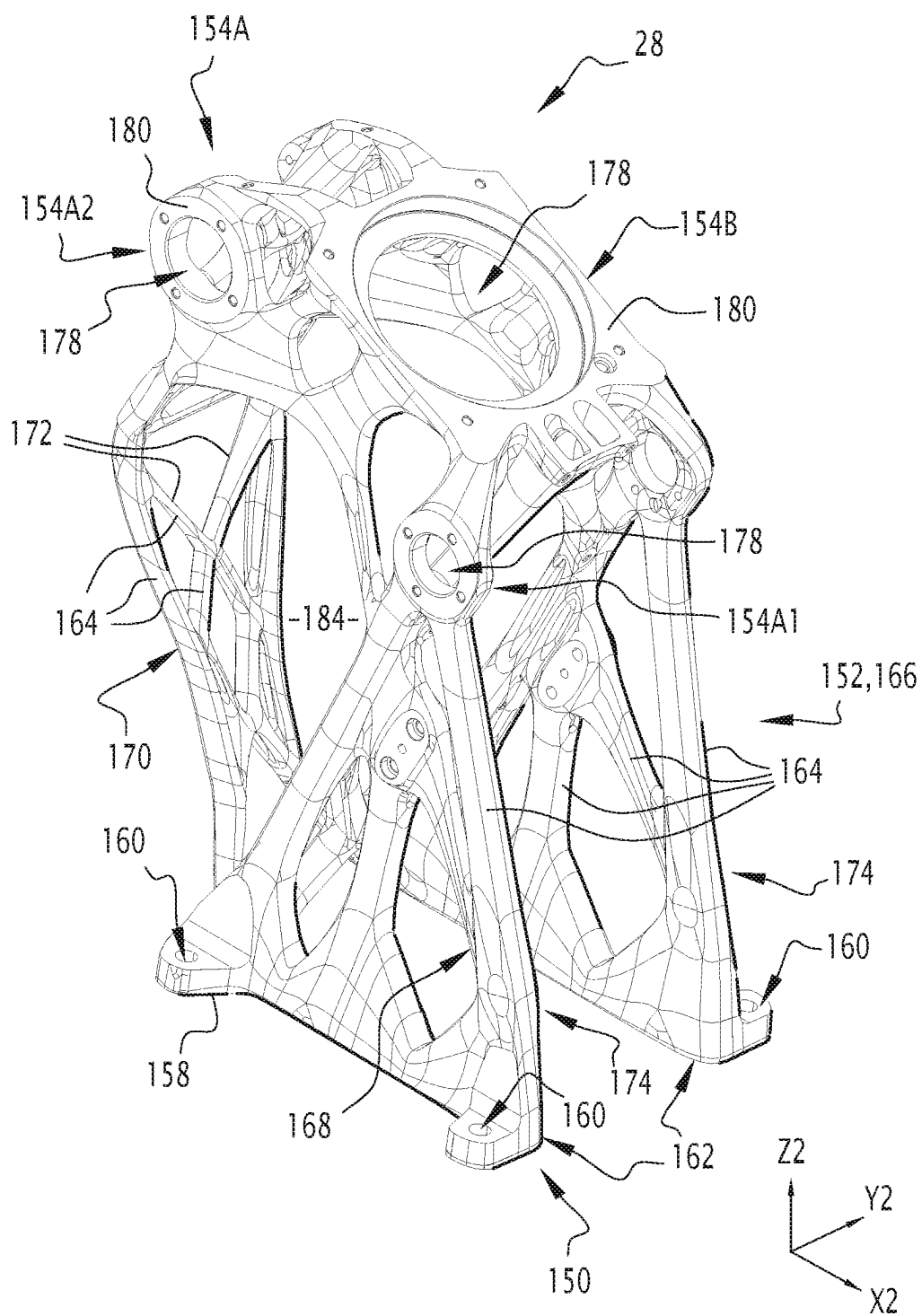


FIG.10

1

AIRCRAFT RUDDER WITH ARTICULATED SUPPORT STRUCTURE OF PEDAL

The present disclosure relates to various embodiments of an aircraft rudder pedal and an aircraft rudder pedal frame. 5

BACKGROUND

In aeronautics, the rudder pedal is one of the main flight controls located in the aircraft cockpit. It usually comprises two pedals that allow the pilot to operate the rudder of the aircraft to control the yaw axis. 10

Some rudder pedals also allow the pilot to control braking of the aircraft. 15

However, the known rudder pedals, and in particular, if need be, the frames of these rudder pedals, do not allow an adaptability of the ergonomics for a wide range of size of the aircraft pilot. 20

Indeed, the adjustment travel of the pedal positions are limited. Moreover, in certain adjusted positions, the pedals can present an inclination and/or a height not suited to the morphology of certain sizes of feet of the pilot. This may cause difficulty for the pilot to reach the foot support area according to the functions sought. 25

SUMMARY

One purpose of the present disclosure is therefore to provide a rudder pedal allowing the pilot, regardless of his size, to control the yaw angle and/or the braking of the aircraft comfortably and safely. 30

Another purpose is to provide a rudder pedal that can be adapted so that the aircraft can be flown by a pilot and/or a co-pilot, to allow each, regardless of their respective sizes, to control the aircraft comfortably and safely while allowing the coupling of yaw control between the pilot and the co-pilot. 35

In addition, it is desirable that each rudder pedal be able to ensure the yaw and/or braking control even in the event of a malfunction of one of the two rudder pedals. 40

To this end, the present disclosure relates to an aircraft rudder pedal comprising a frame and at least one pedal side system, the or each pedal side system comprising a pedal and an articulated pedal support structure connecting the pedal to the frame; wherein the articulated support structure of said pedal comprises a crank, a lever and a support link, the crank presenting a first pivotal articulated connection with the frame and a second pivotal articulated connection with the support link, the lever presenting a first pivotal articulated pivotal connection with the frame and a second pivotal articulated connection with the support link. 45

The rudder pedal according to the present disclosure may comprise one or more of the following features, taken individually or in any technically possible combination: 50

the crank and the lever form a kinematic trapezoid, the straight line passing through the first pivotal articulation connection of the crank with the frame and through the second pivotal articulation connection of the crank with the support link being substantially parallel to the straight line passing through the first pivotal articulation connection of the lever with the frame and through the second pivotal articulation connection of the lever with the support link; 60

the distance between an axis of rotation of the crank relative to the frame and an axis of rotation of the crank relative to the support link is greater than the distance 65

2

between an axis of rotation of the lever relative to the frame and an axis of rotation of the lever relative to the support link;

the rudder pedal comprises two side pedal systems, the articulated support structures of the two side pedal systems being arranged on either side of the frame;

the or each pedal side system comprises a braking system able to control the braking of the aircraft, the braking system comprising the pedal of the pedal side system, said pedal being connected to the crank by a braking pivotal link;

the braking system comprises a braking acquisition system configured to generate an electrical signal representative of a displacement of said pedal relative to the crank about the braking pivotal link, the braking acquisition system being supported by at least one of the crank, the lever, and the support link, preferably by the support link;

the brake acquisition system comprises at least two redundant acquisition sensors, each acquisition sensor comprising a fixed element and a movable element, the movable element being capable of being displaced relative to the fixed element, each acquisition sensor being able to generate an electrical measurement signal as a function of the position of the movable element relative to the fixed element; the braking acquisition system also comprising a device for jointly driving the acquisition sensors, the joint drive device being able to displace, for each acquisition sensor, the movable element relative to the fixed element of the acquisition sensor;

the joint drive device is able to transform a displacement of the pedal relative to the crank about the braking pivotal link into a joint displacement of the movable elements relative to the respective fixed elements;

each acquisition sensor comprises a roller integral with the movable element, the joint drive device comprising a joint drive frame for the rollers of the acquisition sensors, the drive frame being movable to the fixed elements of the acquisition sensors and delimiting, for each roller, a reception housing receiving said roller, the reception housing preferably being a groove, the groove being for example open;

the acquisition sensors are rotatable and the drive frame is adapted to be displaced in rotation relative to the fixed elements of the acquisition sensors;

the drive device further comprises an actuator arm of the drive frame, the actuator arm being able to transform the rotation of the pedal relative to the crank into rotation of the drive frame relative to the fixed elements of the acquisition sensors, the actuator arm being connected on the one hand to the pedal and integral on the other hand with the drive frame;

the actuator arm is articulated and comprises a first drive section connected to the pedal and a second drive section integral with the drive frame, the first drive section presenting a pivotal articulation connection with the second drive section;

each acquisition sensor is able to generate an electrical measurement signal as a function of the position of the movable element relative to the fixed element over a useful electrical measurement travel, the braking acquisition system further comprising a disabling system configured, for each acquisition sensor, to displace the movable element relative to the fixed element out of the useful electrical measurement travel of the acquisition sensor, in the event of uncoupling of the drive device;

3

the disabling system exerts a disabling force on the drive frame, the disabling force being sufficient to displace, for each acquisition sensor, the movable element relative to the fixed element out of the useful electrical measurement travel of the acquisition sensor, in the event of uncoupling of the drive device;

the braking system also comprises a braking force restitution system able to exert a force against a rotation of said pedal relative to the crank about the braking pivotal link, the braking force restitution system being supported by at least one of the crank, the lever and the support rod, preferably by the support rod; and one of the crank, the lever and the support rod comprises a plate and a wall projecting from the plate, the drive frame of the braking force acquisition system being attached to the wall, the braking force restitution system presenting an end integral with the plate.

Furthermore, the present disclosure relates to an aircraft comprising at least one rudder pedal as described above, or at least two rudder pedals as described above.

Alternatively, the present disclosure also has as its object to provide an aircraft measurement acquisition system, suitable for use with any type of aircraft equipment, not limited to a rudder, and allowing good measurement security.

To this end, the present disclosure also relates independently to an aircraft acquisition system comprising at least two redundant acquisition sensors, each acquisition sensor comprising a fixed element and a movable element, the movable element being able to be moved relative to the fixed element, each acquisition sensor being able to generate an electrical measurement signal as a function of the position of the movable element relative to the fixed element over a useful electrical measurement travel, the acquisition system comprising a device for jointly driving the acquisition sensors, the joint drive device being able to displace, for each acquisition sensor, the movable element relative to the fixed element of the acquisition sensor, the acquisition system also comprising a system for disabling the configured travel, for each acquisition sensor, to displace the movable element relative to the fixed element outside the useful electrical measurement travel of the acquisition sensor, in the event of uncoupling of the drive device.

Such an acquisition system is not only adapted to the acquisition of yaw or braking commands, in a rudder pedal, such as described hereafter, but is able to be adapted to any type of aircraft equipment.

The aircraft acquisition system according to the present disclosure may comprise one or more of the following features, taken alone or in any technically feasible combination:

by "joint drive" is meant that the drive device is suitable for simultaneously displacing the movable elements relative to the respective fixed elements by the same relative movement;

the acquisition system comprises at least three redundant acquisition sensors, and for example four redundant acquisition sensors;

each acquisition sensor is a rotary sensor;

each acquisition sensor is an inductive sensor, for example an RVDI sensor (Rotary Variable Differential Transformer), the fixed element then comprising at least one winding, preferably at least one primary winding and one secondary winding, the movable element then comprising a core;

each acquisition sensor is a resistive sensor, for example a potentiometer, the fixed element comprising a resistive track and the movable element comprising a slider;

4

each acquisition sensor comprises a roller integral with the movable element, and the drive device comprises a joint drive frame for the rollers of the acquisition sensors, the drive frame being movable relative to the fixed elements of the acquisition sensors and delimiting, for each roller, a reception housing receiving the roller;

the reception housing is a groove, the groove being, for example, open;

the drive frame comprises a fork for each roller, the fork delimiting the said roller reception housing;

the acquisition sensors are rotatable, and the drive frame is able to be rotated relative to the fixed elements of the acquisition sensors about a predetermined axis of rotation, and, for each acquisition sensor, the rotation of the drive frame, relative to the fixed element, displaces the movable element relative to the fixed element of the acquisition sensor;

the predetermined axis of rotation of the drive frame relative to the fixed elements of the acquisition sensors passes through a geometric center of the drive frame located at the same distance from each roller;

at least two of the rollers are respectively arranged at different distances from said predetermined axis of rotation of the drive frame relative to the fixed elements;

the drive frame is able to be connected to a gripping member of an aircraft control equipment, the gripping member being able to be manipulated by a crew member of the aircraft and being movable relative to another piece of the control equipment;

the gripping member is a pedal, a handle, or a stick;

the drive frame is mechanically connected to the gripping member;

the disabling system exerts a disabling force on the drive frame, the disabling force being sufficient to displace each acquisition sensor out of the useful measurement range of the acquisition sensor, in case of uncoupling of the drive device;

the disabling system comprises at least one spring or set of springs able to exert the disabling force;

the drive device further comprises an actuator arm able to displace the drive frame relative to the fixed elements of the acquisition sensors in order to generate each electrical measurement signal, the drive frame being able to be connected to the gripping member by means of the actuator arm;

the actuator arm is able to transform a displacement of the gripping member relative to said other piece of control equipment into rotation of the drive frame relative to the fixed elements of the acquisition sensors around the predetermined axis of rotation; and

the actuator arm exerts, in the absence of uncoupling of the drive device, a holding force on the drive frame opposite to the disabling force, the holding force being greater than or equal to the disabling force

"uncoupling of the drive device" means any event from which the drive device is no longer able to move the movable elements together relative to the respective fixed elements;

for example, it means any failure, blockage or breakage of a part of the joint drive or of a connection between two parts of the drive, the term "breakage" designates, in particular, the fracture of a solid object into two or more parts under excessive stress or strain; and

for example, it is also means any fault in the assembly of one or more parts of the drive device, such as, for

5

example the omission of a fixing screw, the loosening of one of the screws by vibration, or the misalignment of parts.

The present disclosure also relates to a device for controlling an aircraft piloting or flight parameter comprising an acquisition system as described above and a processing unit, the processing unit being configured:

- to receive the measurement signals generated in parallel by the acquisition sensors of the acquisition system,
- to verify that each of the parallel measurement signals belongs to the useful electrical measurement travel of the acquisition sensors, and
- to elaborate a control signal for the control or flight parameter from the parallel measurement signals verified as belonging to the useful electrical measurement travel of the acquisition sensors.

The control signal for the control or flight parameter is therefore not generated from signals that do not belong to the useful electrical measurement travel of the acquisition sensors.

By “parallel generated measurement signals” or “parallel measurement signals” is meant signals generated by each of the redundant acquisition sensors for the same relative displacement of the movable elements by the drive device.

The control device preferably further includes control equipment comprising a gripping member and at least one other part, the gripping member being able to be manipulated by a crew member of the aircraft and being movable relative to the other part, the joint drive device being able to transform a displacement of the gripping member relative to said other part of the control equipment into a joint displacement of the movable elements relative to the respective fixed elements.

In particular, the drive frame is, for example, mechanically connected to the gripping member.

The gripping member is a pedal, a handle, or a stick.

The control or flight parameter is, for example, a yaw angle, a braking, a roll angle, a pitch angle, a heading, a trajectory, an altitude, a thrust of at least one engine of the aircraft, an air speed, a ground speed, a rate of climb, a rate of descent, or an acceleration.

The present disclosure also relates to an aircraft comprising a control device as described above.

BRIEF SUMMARY OF THE DRAWINGS

The present disclosure will be better understood upon reading the following description, given only by way of example, and made with reference to the attached drawing, in which:

FIG. 1 is a schematic view of an example of an aircraft according to the present disclosure;

FIG. 2 is a perspective schematic view of an example of a control device of the aircraft of FIG. 1;

FIGS. 3 and 4 are front and side schematic views, respectively, of a rudder pedal of the control device of FIG. 2;

FIGS. 5 and 6 are perspective schematic sections of the rudder pedal;

FIG. 7 is a perspective schematic view of a braking system of the rudder pedal;

FIG. 8 is a schematic perspective section of the braking system of FIG. 7;

FIGS. 9 and 10 are schematic perspective views of the rudder pedal frame alone.

6

DETAILED DESCRIPTION

An example of an aircraft 10 is illustrated in FIG. 1.

The aircraft 10 includes at least one rudder 12, suitable for producing a yawing motion of the aircraft 10.

The aircraft 10 also comprises at least one landing gear 14 of the aircraft 10, the landing gear 14 comprising wheels and brakes suitable for braking said wheels.

Here and for the following, is defined:

- a longitudinal direction X1 which is parallel to a longitudinal axis L of the aircraft 10;
- a vertical direction Z1 which forms with the longitudinal direction X1 a vertical plane which is parallel to a vertical plane of symmetry of the aircraft 10, the vertical direction Z1 being orthogonal to the longitudinal direction X1; and
- a lateral direction Y1 that is orthogonal to said longitudinal direction X1 and vertical direction Z1.

The terms “rear” and “front” will then be understood relative to the longitudinal direction X1, namely for “front” toward the front of the aircraft 10 in the direction of a flight of the aircraft 10, and for “rear” toward the rear of the aircraft 10 in the direction opposite to the direction of a flight of the aircraft 10.

The aircraft 10 comprises a cockpit 16 preferably intended to accommodate at least two pilots. In particular, the cockpit 16 comprises a pilot seat for each pilot.

The aircraft 10 includes, for example, a display system comprising a screen, the screen preferably being arranged in the cockpit 16 for the pilots.

The aircraft 10 comprises a control device 20 illustrated in particular in FIG. 2.

The control device 20 comprises in this example at least two rudder pedals 22, and a main connecting rod 24 connecting the two rudder pedals 22.

The control device 20 preferably also comprises a processing unit 26 visible in FIG. 1, the processing unit 26 being able to implement the yaw angle and/or braking control functions.

Indeed, as will be described in more detail below, each rudder pedal 22 is preferably suitable to be actuated to allow the pilot to control the yaw angle of the aircraft 10 during a flight, in particular by controlling the rudder 12. In addition, each rudder pedal 22 is advantageously able to be actuated to allow the pilot to control the braking of the aircraft 10, when the wheels of the aircraft 10 are in contact with the ground, in particular by controlling the brakes of the landing gear 14.

Each rudder pedal 22 of the control device 20 is arranged in the cockpit 16 of the aircraft 10. In particular, each rudder pedal 22 is arranged opposite one of the seats so that the pilot seated in the seat can actuate the rudder pedal 22.

Each rudder pedal 22 comprises a frame 28, a preferred non-limiting embodiment of which will be described in more detail relative to FIGS. 9 and 10.

Each rudder pedal 22 comprises at least one pedal side system 30, each pedal side system 30 comprises a pedal 32 and an articulated support structure 34 connecting the pedal 32 to the frame 28.

Each rudder pedal 22 also comprises an output shaft 36, and a mechanical kinematic chain 38 for transmitting to the output shaft 36 a displacement of the pedals 32 relative to the frame 28, to transmit a yaw command from the pilot to the output shaft 36.

The output shafts 36 of the two rudders 22 are connected to each other via the main connecting rod 24.

Advantageously, each rudder pedal **22** also comprises a yaw acquisition system **40**.

In addition, each rudder pedal **22** preferably comprises a yaw force restitution system **42**.

Advantageously, each rudder pedal **22** also comprises an ergonomic adjustment system **44** for adjusting an operating position of the pedals **32** relative to the frame **28**.

As shown above, each pedal side system **30** comprises the pedal **32** and the pedal articulated support structure **34** of the pedal **32**.

As illustrated in FIGS. **2** to **4**, each rudder pedal **22** preferably comprises two pedal side systems **30**. The articulated support structures **34** of the two pedal side systems **30** are then arranged on one side or the other of the frame **28**.

Each pedal side system **30** will now be described in more detail.

The pedal **32** comprises at least one rest **46** for a foot of the pilot.

The pedal **32** has an outer contour surrounding the footrest **46**.

As illustrated in FIG. **3**, the footrest **46** is, for example, solid, whereby the surface area of the foot rest **46** corresponds to the entire area delimited by said outer contour. Alternatively, the footrest **46** is perforated.

As illustrated in FIGS. **3** to **5**, the articulated support structure **34** of the pedal **32** includes at least one crank **48**.

The displacement of the pedal **32** relative to the frame **28** is permitted at least by said crank **48**.

Thus, the crank **48** presents on the one hand a first pivotal articulation connection **50** with the frame **28** and is connected on the other hand to the pedal **32**. The connection of the crank **48** to the pedal **32** is preferably a braking pivot connection, as described in more detail below.

By "pivotal articulation connection between two members" is meant here and hereafter any system allowing freedom of rotation between the two members according to a single axis. In a non-limiting manner, such a system comprises, for example, a straight axis part around which one or more other parts of the system rotate, the straight axis part defining the axis of rotation of the connection. Such a system also additionally or alternatively comprises at least one roller bearing and/or at least one bearing.

By means of the first pivotal articulation connection **50**, the crank **48** is able to be rotated relative to the frame **28** by a displacement of said pedal **32** relative to the frame **28**. The rotation is then according to an axis of rotation **A1** passing through the first pivotal articulation connection **50** of the crank **48** with the frame **28**.

In particular, when the pilot presses on the pedal **32** to the right of the connection of the crank **48** to the pedal **32**, the displacement of the pedal **32** jointly drives the crank **48** in rotation relative to the frame **28** according to the axis of rotation **A1**.

Such a joint displacement of the pedal **32** and the crank **48** relative to the frame **28** according to the axis of rotation **A1** corresponds, for example, to a rotation of the foot of the pilot about his knee.

Such a joint displacement of the pedal **32** and the crank **48** relative to the frame **28** according to the axis of rotation **A1** is intended to control the yaw angle, by means of the mechanical kinematic chain **38**, the yaw acquisition system **40**, and the processing unit **26** as explained in more detail later.

In the example of FIGS. **3** to **5**, the crank **48** extends, between the first pivotal articulation connection **50** and the connection to the pedal **32**, along a guide curve. Preferably, the guide curve is straight, when viewed projected onto a

plane perpendicular to the axis of rotation **A1**. Alternatively, this guide curve is curved when viewed projected on a plane perpendicular to the axis of rotation **A1**, the crank **48** being, in other words, curved.

In one preferred embodiment of the present disclosure, the articulated support structure **34** of the pedal **32** further comprises a lever **52** and a support link **54**.

The displacement of the pedal **32** relative to the frame **28** about the axis of rotation **A1** is then allowed, at least by the assembly formed by the crank **48**, the support link **54** and the lever **52**, when the pilot presses the pedal **32**.

In this example, the crank **48** presents a second pivotal articulation connection **56** with the support link **54**.

In other words, the crank **48** is able to be rotated relative to the support link **54** about an axis of rotation **A2** passing through the second pivotal articulation connection **56** of the crank **48** with the support link **54**.

The axis of rotation **A1** of the crank **48** relative to the frame **28** is parallel to the axis of rotation **A2** of the crank **48** relative to the support link **54**.

The distance between the axis of rotation **A1** of the crank **48** relative to the frame **28** and the axis of rotation **A2** of the crank **48** relative to the support link **54** remains constant during any displacement of the pedal **32** relative to the frame **28**. For example, the crank **48** is thus rigid and non-deformable.

The lever **52** presents a first pivotal articulation connection **58** with the frame **28** and a second pivotal articulation connection **60** with the support link **54**.

In other words, the lever **52** is able to be rotated relative to the frame **28** about an axis of rotation **A3** passing through the first pivotal articulation connection **58** of the lever **52** with the frame **28**. In addition, the lever **52** is able to be rotated relative to the support link **54** about an axis of rotation **A4** passing through the second pivotal articulation connection **60** of the lever **52** with the support link **54**.

The axis of rotation **A3** of the lever **52** relative to the frame **28** is parallel to the axis of rotation **A4** of the lever **52** relative to the support link **54**.

Furthermore, the axis of rotation **A3** of the lever **52** relative to the frame **28** is parallel to the axis of rotation **A1** of the crank **48** relative to the frame **28**.

The distance between the axis of rotation **A3** of the lever **52** relative to the frame **28** and the axis of rotation **A4** of the lever **52** relative to the support link **54** remains constant during any displacement of the pedal **32** relative to the frame **28**. For example, the lever **52** is thus rigid and non-deformable.

In the example shown in FIG. **4**, the lever **52** extends, between the first pivotal articulation connection **58** with the frame **28** and the second pivotal articulation connection **60** with the support link **54**, along a guiding curve. Preferably, the guiding curve is straight, when viewed projected onto a plane perpendicular to the axis of rotation **A1**. Alternatively, this guiding curve is curved when projected onto a plane perpendicular to the axis of rotation **A1**, the lever **52** being, in other words, curved.

The support link **54** extends at least from the crank **48** to the lever **52**.

The axis of rotation **A4** of the lever **52** relative to the support link **54** is parallel to the axis of rotation **A2** of the crank **48** relative to the support link **54**.

The distance between the axis of rotation **A2** of the crank **48** relative to the support link **54** and the axis of rotation **A4** of the lever **52** relative to the support link **54** remains

constant during any displacement of the pedal 32 relative to the frame 28. For example, the support link 54 is thus rigid and non-deformable.

Thus, when the pilot presses the pedal 32 to the right of the connection of the crank 48 to the pedal 32, the displacement of the pedal 32 jointly drives the crank 48 in rotation relative to the frame 28 according to the axis of rotation A1, as well as the lever 52 in rotation relative to the frame 28 according to the axis of rotation A3, by means of the support link 54.

Preferably, the crank 48 and the lever 52 form a kinematic trapezoid.

More specifically, the straight line passing through the first pivotal articulation connection 50 of the crank 48 with the frame 28 and through the second pivotal articulation connection 56 of the crank 48 with the support link 54 is substantially parallel to the straight line passing through the first pivotal articulation connection 58 of the lever 52 with the frame 28 and through the second pivotal articulation connection 60 of the lever 52 with the support link 54.

A "straight line through the pivotal articulation connection" is defined as a straight line through the axis of rotation associated with this pivotal articulation connection.

This parallelism is maintained during any displacement of the pedal 32 relative to the frame 28.

In the preferred example, the distance between the axis of rotation A1 of the crank 48 relative to the frame 28 and the axis of rotation A2 of the crank 48 relative to the support link 54 is different from the distance between the axis of rotation A3 of the lever 52 relative to the frame 28 and the axis of rotation A4 of the lever 52 relative to the support link 54.

Specifically, the distance between the axis of rotation A1 and the axis of rotation A2 is greater than the distance between the axis of rotation A3 and the axis of rotation A4, for example, at least 5 mm greater than the distance between the axis of rotation A3 and the axis of rotation A4.

As will be explained later, the use of the kinematic trapezoid with a difference in length between the crank 48 and the lever 52 allows for a slight variation in the angle of the pedal when ergonomically adjusting the rudder pedal. It results in a good ergonomic evolution of the angle imposed on the ankle of the pilot. This evolution of the angle responds to an ergonomic constraint which aims at not reaching extreme angles of the ankle which are uncomfortable for the pilot.

In the illustrated example, the articulated support structure 34 of the pedal 32 has no other crank or lever connected to the frame 28 and allows the rotation of the pedal 32 relative to the frame 28 around the axis of rotation A1, in other words, moving together with the pedal 32 during such a rotation.

The mechanical kinematic chain 38 for transmitting a displacement of the pedals 32 relative to the frame 28 will now be described.

By "mechanical kinematic chain" is meant an assembly of parts connected to each other, the assembly being able to transmit and/or transform a movement, the parts comprising, for example, at least those described below.

Generally speaking, as illustrated in FIGS. 4 to 6, the mechanical kinematic chain 38 comprises at least one central transmission part 62.

The mechanical kinematic chain 38 comprises, for each pedal 32, a transformation mechanism 64 for a displacement of the pedal 32 relative to the frame 28 into rotation of the central transmission part 62 relative to the frame 28.

The mechanical kinematic chain 38 also comprises a transmission mechanism 66 joining the central transmission part 62 to the output shaft 36.

The central transmission part 62 is able to be rotated, according to an axis of rotation A5, relative to the frame 28 by a displacement of the pedals 32 relative to the frame 28, this displacement being here the rotation of the pedals 32 relative to the frame 28 about the respective axes A1.

The central transmission part 62 is able to be rotated relative to the frame 28 according to the axis A5 by the two transformation mechanisms 64 that will now be described.

Any mechanism could be suitable as long as the central transmission part 62 can be rotated relative to the frame 28 according to the axis A5 by a displacement of the pedals 32 relative to the frame 28 about the respective axes A1.

The transformation mechanisms 64 are preferably symmetrical the one from the other relative to a median plane of the frame 28. The median plane of the frame 28 here passes through the longitudinal direction X1 and the vertical direction Z1.

In a preferred example of one embodiment, for each pedal 32, the transformation mechanism 64 comprises at least said crank 48 of the pedal side system 30 and an intermediate link 68.

The intermediate link 68 in this example ensures the connection between the central transmission part 62 and the crank 48.

Thus, the intermediate link 68 presents a first point of articulation 70 with the crank 48 and a second point of articulation 72 with the central transmission part 62.

Here and hereafter, each articulation point is a pivotal articulation connection as defined above or a ball and socket connection.

In other words, the intermediate link 68 is able to be rotated relative to the crank 48 about an axis of rotation A6 passing through the first point of articulation 70 of the intermediate link 68 with the crank 48. In addition, the intermediate link 68 is able to be rotated relative to the central transmission part 62 about an axis of rotation A7 passing through the second point of articulation 72 of the intermediate link 68 with the central transmission part 62.

The distance between the axis of rotation A6 of the intermediate link 68 relative to the crank 48 and the axis of rotation A7 of the intermediate link 68 relative to the central transmission part 62 remains constant during any displacement of the pedal 32 relative to the frame 28.

For example, the intermediate link 68 is thus rigid and non-deformable. The intermediate link is preferably one-piece.

The first point of articulation 70 of the intermediate link 68 with the crank 48 is away from the first pivotal articulation connection 50 of the crank 48 with the frame 28, and preferably away from the connection of the crank 48 with the pedal 32.

In particular, as illustrated in FIG. 4, the first point of articulation 70 of the intermediate link 68 with the crank 48 is arranged between the first pivotal articulation connection 50 of the crank 48 with the frame 28 and the connection of the crank 48 with the pedal 32.

In the example shown in FIG. 4, the axis of rotation A6 passing through the first point of articulation 70 of the intermediate link 68 with the crank 48 extends non-parallel to the axis of rotation A7 passing through the second point of articulation 72 of the intermediate link 68 with the central transmission part 62.

Preferably, the axis of rotation A7 passing through the second point of articulation 72 of the intermediate link 68

11

with the central transmission part 62 is parallel to the axis of rotation A5 of the central transmission part 62 relative to the frame 28.

For example, at the second point of articulation 72, the central transmission part 62 comprises a fork and the intermediate link 68 presents a connecting end. The fork and the connecting end then form said second point of articulation 72 of the intermediate link 68 with the central transmission part 62.

Thus, when the crank 48 of the transformation mechanism 64 is rotated relative to the frame 28 according to the axis A1, the rotation of the crank 48 moves the intermediate link 68 of the transformation mechanism 64 relative to the frame 28, and the displacement of the intermediate link 68 rotates the central transmission part 62 relative to the frame 28 according to the axis A5.

This rotation of the central transmission part 62 relative to the frame 28 according to the axis A5 is driven whether the pilot presses on one or the other of the pedals 32.

In one preferred embodiment, the mechanical kinematic chain 38 is able to generate displacements of the pedals 32 relative to the frame 28, according to the respective axes A1, which are antagonistic to each other.

In other words, the mechanical kinematic chain 38 is such that the rotation of one of the pedals 32 relative to the frame 28 according to a first direction of rotation about its axis A1 causes an opposite rotation of the other of the pedals 32 relative to the frame 28 in a second direction of rotation opposite to the first direction of rotation, when the directions of rotation are seen from the same side.

Indeed, the rotation of the central transmission part 62 according to the axis of rotation A5, caused by one of the transformation mechanisms 64, drives the other of the transformation mechanisms 64 and therefore the associated pedal 32 in an opposite manner.

Thus, the two intermediate links 68 are located on either side of the central transmission part 62.

More specifically, the rotation of the central piece relative to the frame 28 according to the axis of rotation A5, caused by the intermediate link 68 of one of the transformation mechanisms 64, antagonistically displaces the intermediate link 68 of the other of the transformation mechanisms 64, and therefore the associated crank 48 and pedal 32.

The mechanical kinematic chain 38 is able to transmit, to the output shaft 36, the rotation of the central transmission part 62 relative to the frame 28 according to the axis of rotation A5, this by means of the transmission mechanism 66.

The transmission mechanism 66 comprises at least one homokinetic association of two universal joints 74 between said central transmission part 62 and the output shaft 36.

By "universal joint" is meant a connection system that allows the transmission of angular rotation from one part to another part, the axes of rotation of the parts being concurrent.

More precisely, the output shaft 36 is able to be rotated relative to the frame 28 according to an axis of rotation A8 parallel to the axis of rotation A5 of the central part relative to the frame 28.

By the homokinetic association of the two universal joints 74, the speed of rotation of the central transmission part 62 relative to the frame 28 according to the axis of rotation A5 is equal, at each instant, to the speed of rotation of the output shaft 36 relative to the frame 28 according to the axis of rotation A8.

Thus, when the crank 48 of the transformation mechanism 64 is rotated relative to the frame 28 according to the axis

12

A1, the rotation of the crank 48 displaces the intermediate link 68 of the transformation mechanism 64 relative to the frame 28, the displacement of the intermediate link 68 rotates the central transmission part 62 relative to the frame 28 according to the axis A5, and the transmission mechanism 66 transmits the rotation of the central part to the output shaft 36 relative to the frame 28 according to the axis A8.

The two rudder pedals 22 are connected to each other by means of the main connecting rod 24 such that the relative positions of the pedals 32 of one of the rudder pedals 22 simultaneously reflect the relative positions of the pedals 32 of the other of the rudder pedals 22.

Generally, the main connecting rod 24 is arranged above the pedals 32, and above the frame 28, projecting in the vertical direction Z1. In addition, the main connecting rod 24 is arranged in a proximal manner relative to the rudder pedal 22.

The mounting of the main connecting rod 24 is thus facilitated by its particular arrangement.

In the preferred embodiment, the connection of the output shafts 36 by the main connecting rod 24 is such that a displacement of the pedals 32 relative to the frame 28 of one of the rudder pedals 22 is transmitted to the pedals 32 of the other of the rudder pedals 22.

To this end, in the embodiment illustrated in FIG. 2, the main connecting rod 24 is able to transmit the rotation of the output shaft 36 relative to the frame 28 according to the axis A8 of one of the rudder pedals 22 into rotation, in the same direction, of the output shaft 36 relative to the frame 28 according to the axis A8 of the other of the rudder pedals 22.

For this purpose, the main connecting rod 24 is connected on both sides to the output shafts 36 of both rudder pedals 22.

Preferably, the main connecting rod 24 presents a pivotal articulation connection 76 to each output shaft 36 of the rudder pedals 22.

In other words, for each output shaft 36 of the rudder pedals 22, the main connecting rod 24 is able to be rotated relative to the output shaft 36 about an axis of transmission A14 passing through the pivotal articulation connection 76 of the main connecting rod 24 with that output shaft 36.

In the example in FIG. 2, the two axes of transmission A14 are parallel to each other.

The axes of transmission A14 are parallel to the respective axes of rotation A9 of the output shafts 36 relative to the frames 28.

Thus, when one of the pilots presses one of the pedals 32 of a first rudder pedal 22, the joint rotation of the pedal 32 and the crank 48 relative to the frame 28 according to the axis A1 is transformed by the first rudder pedal 22 into rotation of the output shaft 36 relative to the frame 28 according to the axis A8 of this first rudder pedal 22, the main connecting rod 24 transmits this rotation into rotation of the output shaft 36 relative to the frame 28 according to the axis A8 of the second rudder pedal 22, which is transformed by the second rudder pedal 22 into rotation of the pedals 32 relative to the frame 28 of this second rudder pedal 22.

In parallel with the connection of the two rudder pedals 22 by the output shafts 36 and the main connecting rod 24, it is also possible for each rudder pedal 22 to independently measure a yaw angle command from the rotation of the output shaft 36 relative to the frame 28 according to the axis A8, and to exert an artificial force against the displacement of the pedals 32 relative to the frame 28 in order to restore a force related to the yaw angle command.

13

The yaw acquisition system **40** is supported by the frame **28**.

Thus, each rudder pedal **22** carries its own yaw acquisition system **40**, which allows a redundancy of the yaw control, and to avoid depending on a central acquisition station common to both rudder pedals **22**. The safety is improved.

The yaw acquisition system **40** is configured to generate an electrical signal representative of the displacement of the pedals **32** relative to the frame **28**.

In the illustrated embodiment, the yaw acquisition system **40** is configured to generate an electrical signal representative of the rotation of the pedals **32** relative to the frame **28**, according to the respective axes of rotation **A1**.

Generally, the yaw acquisition system **40** is configured to generate said signal from a measurement of displacement of any of the parts of the mechanical kinematic chain **38** relative to the frame **28**.

In the illustrated example, the yaw acquisition system **40** is in particular configured to generate said electrical signal from a measurement of the rotation of the output shaft **36** relative to the frame **28** according to the axis of rotation **A8**.

To do so, the yaw acquisition system **40** comprises at least one acquisition sensor **78**.

In the preferred embodiment shown in FIG. 5, the yaw acquisition system **40** comprises at least two redundant acquisition sensors **78**, and a joint drive device **80** for the acquisition sensors **78**.

Preferably, the yaw acquisition system **40** further comprises a system **82** for disabling the acquisition sensors **78** that presents the same features as those described below for the braking acquisition system **112**.

Each acquisition sensor **78** comprises a fixed element and a movable element, the movable element being able to be displaced relative to the fixed element.

Each fixed element is fixed relative to the frame **28**.

In particular, each acquisition sensor **78** comprises a roller **84** integral with the movable element.

The acquisition sensors **78** are preferably rotatable.

Each acquisition sensor **78** is able to generate an electrical measurement signal as a function of the position of the movable element relative to the fixed element over a useful electrical measurement travel.

The joint drive device **80** is able to displace, for each acquisition sensor **78**, the movable element relative to the fixed element of the acquisition sensor **78**.

The acquisition sensors **78** are advantageously resistive sensors, for example potentiometers, the fixed element then comprising a resistive track and the movable element then comprising a slider.

Throughout the following, the term "potentiometer" will be used to refer to each acquisition sensor **78** of the yaw acquisition system **40**, the term "track" will be used to refer to the fixed element of each sensor **78**, and the term "slider" will be used to refer to the movable element of each sensor **78**. However, it is understood that the acquisition sensors **78** may be any type of sensor, other than a potentiometer, preferably any type of rotary sensor, for example any type of resistive or inductive sensor.

The joint drive device **80** is able to simultaneously move the sliders relative to the respective tracks by the same relative displacement.

The joint drive device **80** is able to transform a displacement of the output shaft **36** relative to the frame **28** according to the axis **A8** into a joint displacement of the sliders relative to the respective tracks.

14

For this purpose, the joint drive device **80** preferably comprises a joint drive frame **86** for the rollers **84** of the potentiometers **78**.

The drive frame **86** is then movable relative to the tracks of the potentiometers **78** and delimits, for each roller **84**, a reception housing receiving the roller **84**.

The reception housing is here an open groove.

In particular, the drive frame **86** comprises a fork for each roller **84**, the fork delimiting said roller **84** reception housing.

In the preferred example where the potentiometers **78** are rotatable, the drive frame **86** is able to be displaced in rotation relative to the tracks of the potentiometers **78** about a predetermined axis of rotation.

The predetermined axis of rotation of the drive frame **86** relative to the tracks passes through a geometric center of the drive frame **86**, the geometric center being located at the same distance from each roller **84**.

The drive frame **86** is able to be displaced in rotation relative to the tracks in conjunction with the rotation of the output shaft **36** relative to the frame **28** according to the axis **A8**.

Preferably, the drive frame **86** is integral with the output shaft **36**.

Thus, upon rotation of the output shaft **36** relative to the frame **28** according to the axis **A8**, the output shaft **36** drives in rotation the drive frame **86** and therefore each slider relative to the associated track.

For each redundant potentiometer **78**, an electrical measurement signal is generated in parallel.

By "parallel generated measurement signals" or "parallel measurement signals" is meant signals generated by each of the redundant potentiometers **78** for the same relative displacement of the sliders by the joint drive device **80**.

A yaw angle control can then be developed from these parallel measurement signals.

In the preferred embodiment, the processing unit **26** of the control device **20** is able to implement the yaw angle control function.

The processing unit **26** comprises, for example, a computer processing device and a memory.

The computer processing device is operatively connected to the memory.

The computer processing device corresponds to, for example, a digital signal processor (DSP), a microcontroller, a programmable cell array (FPGA), and/or a dedicated integrated circuit (ASIC) able to perform various data processing operations and functions.

For example, the computer processing device comprises a single processor. Alternatively, the computer processing device comprises several processors, which are located in the same geographical area, or are, at least partially, located in different geographical areas and are then able to communicate with each other.

By the term "memory" is meant any volatile or non-volatile computer memory appropriate to the presently disclosed subject matter, such as random access memory (RAM), read-only memory (ROM), or other electronic, optical, magnetic, or other computer-readable storage medium on which the data and control functions as described herein are stored.

Therefore, the memory is a tangible storage medium where the data and control functions are stored in a non-transitory form.

The processing unit **26** is connected to the yaw acquisition system **40** and to the or each rudder **12** of the aircraft **10**.

15

For ease of clarity, the various wiring, particularly electrical, has not been illustrated in the figures.

The processing unit 26 is configured to receive each measurement signal from the yaw acquisition system 40.

More specifically, in the preferred example embodiment, the processing unit 26 is configured to receive the measurement signals generated in parallel by the acquisition sensors 78 of the yaw acquisition system 40.

The processing unit 26 is configured to generate a yaw angle control signal at least from each measurement signal received from the yaw acquisition system 40, and in particular from the parallel measurement signals.

The generation is implemented from, for example, a yaw control law, stored in the memory of the processing unit 26.

The processing unit 26 is then configured to send the developed yaw angle control signal to the or each rudder 12 of the aircraft 10.

The yaw force restitution system 42 is able to exert an opposing force against a pilot-initiated displacement of the pedals 32 relative to the frame 28 according to their respective axes A1.

The yaw force restitution system 42 is supported by the frame 28.

Generally, the yaw force restitution system 42 is configured to exert said opposing force against the displacement of any one of the parts of the mechanical kinematic chain 38 relative to the frame 28.

The yaw force restitution system 42 defines a rest position of the pedals 32 relative to the frame 28 according to their respective axes of rotation A1, the rest position being that adopted by the pedals 32 in the absence of force exerted by the pilot.

In a preferred embodiment, the opposing force exerted by the yaw force restitution system 42 presents a behavioral law that is at least proportional to the deflection of the pedals relative to the rest position, the coefficient of proportionality corresponding to an equivalent stiffness of the yaw force restitution system 42.

In a preferred embodiment, the opposing force exerted by the yaw force restitution system 42 is preferably double-sloped.

The equivalent stiffness is then a function of the pedal travel relative to the rest position. The equivalent stiffness presents at least a first constant value, until the pedals reach a predetermined travel, and a second distinct constant value beyond the predetermined travel.

The second value is, for example, strictly less than the first value.

In the illustrated embodiment, the yaw force restitution system 42 comprises at least one force member and a force shaft.

The force member comprises, for example, a spring.

The force member is able to exert said opposing force on the force shaft, the force shaft being joined to any one of the parts of the mechanical kinematic chain 38.

In the preferred embodiment illustrated, the yaw force restitution system 42 is in particular configured to exert said opposing force against the rotation of the output shaft 36 relative to the frame 28 according to the axis A8.

The force shaft is then, for example, joined to one of the two universal joints 74 of the mechanical kinematic chain 38.

In the example in FIG. 5, the force shaft is formed by the output shaft 36.

In particular, the output shaft 36 here extends on either side of the associated universal joint 74, with one side of the

16

output shaft 36 forming the force shaft, and the other side being connected to the main connecting rod 24.

The ergonomic adjustment system 44 for an operating position of the pedals 32 relative to the frame 28 will now be described.

The term "pedal operating position" is defined, for example, by the position of the connection point of each pedal 32 to the associated crank 48 in projection onto a median plane of the frame 28, when the pedals 32 are arranged symmetrically relative to the median plane of the frame 28. Indeed, when the pedals 32 are thus arranged symmetrically in this manner, the two connection points of each pedal 32 to the crank 48 are superimposed in projection in the median plane of the frame 28.

Each of these connection points corresponds substantially to the position of the heel of one of the feet of the pilot in space.

The position of use of the pedals 32 relative to the frame 28 preferably corresponds to the rest position defined by the yaw force restitution system 42.

The ergonomic adjustment system 44 is able to move the operating position of the pedals 32 forward or backward relative to the seat of the pilot. The ergonomic adjustment system 44 is thus able to provide the most ergonomic operating position of the pedals 32 for any type of pilot.

The ergonomic adjustment system 44 thus defines an extreme distal operating position and an extreme proximal operating position between which the operating position of the pedals 32 can be locked.

Here and hereafter, the terms "distal" and "proximal" will be understood relative to the pilot operating the rudder pedal. Specifically, a "proximal" element is understood to be closer to the pilot than a "distal" element.

In particular, when the rudder pedal 22 is installed in the aircraft 10, then the terms "distal" and "proximal" are synonymous with "front" and "back" respectively.

The adjustment travel allowed by the ergonomic adjustment system 44 is at least 100 mm. In particular, the adjustment travel is the distance between the distal end use position and the proximal end use position.

In general, the ergonomic adjustment system 44 is able to jointly displace each pedal 32, in particular for moving the operating position forward or backward relative to the seat of the pilot.

More specifically, the ergonomic adjustment system 44 is able to jointly displace each pedal 32 in a plane perpendicular to the respective axes of rotation A1 of the cranks 48 relative to the frame 28.

To this end, in one preferred embodiment, the ergonomic adjustment system 44 comprises a slide 88 fixed relative to the frame 28 and a carriage 90.

Preferably, the ergonomic adjustment system 44 also comprises a device 92 for displacing and locking the carriage 90 to the slide 88.

The carriage 90 is able to be displaced on the slide 88 along an adjustment direction and is able to be locked in position on the slide 88.

The adjustment direction is inscribed in a plane passing through the longitudinal direction X1 and the vertical direction Z1. This plane is in particular a median plane of the frame 28.

As illustrated in FIGS. 4 and 5, the central transmission part 62 is carried by the carriage 90. Preferably, the central transmission part 62 surrounds the carriage 90.

The two intermediate links 68 are arranged on either side of the carriage 90.

17

The axis of rotation A5 of the central transmission part 62 is perpendicular to the direction of adjustment of the carriage 90.

Thus, the central transmission part 62 is able to be rotated according to the axis of rotation A5 relative to the carriage 90, by displacement of the pedals 32 relative to the frame 28.

In other words, throughout the foregoing, when reference is made to rotation of the central transmission part 62 relative to the frame 28 according to the axis of rotation A5, this is synonymous with rotation of the central part 62 relative to the carriage 90 according to the axis A5, when the carriage 90 is locked in position relative to the slide 88.

The slide 88 comprises at least one guide rod 94 extending according to the adjustment direction. In the example shown in FIG. 8, the slide 88 comprises two guide rods 94 extending according to the adjustment direction. Alternatively, the slide 88 comprises a single guide rod 94 or more than two guide rods 94.

In particular, the carriage 90 presents, for each guide rod 94, a guide hole receiving the guide rod 94.

A displacement of the carriage 90 relative to the slide 88 simultaneously causes a displacement of the central transmission part 62 relative to the frame 28 in the adjustment direction and thus a displacement of the operating position of the pedals 32 relative to the frame 28.

Indeed, when the carriage 90 is displaced relative to the slide 88 in the adjustment direction, the displacement of the central transmission part 62 displaces each intermediate link 68 relative to the frame 28, and the displacement of each intermediate link 68 drives in rotation the associated crank 48 about the axis of rotation A1 relative to the frame 28 and thus the pedals 32. The two pedals 32 are thus jointly displaced by the same distance. They are moved closer to or further away from the seat of the pilot by this same distance.

The ergonomic adjustment of the pedals 32 of the rudder pedal 22 does not cause any rotation of the central transmission part 62 relative to the frame 28 about the axis of rotation A5.

Thus, despite the connection of the output shafts 36 of the two rudder pedals 22 by the main connecting rod 24, the ergonomic adjustments of the rudder pedals 22 are independent of each other.

This is clearly shown in FIG. 2, which shows separate ergonomic settings for the two rudder pedals 22.

During the ergonomic adjustment, the angular orientation of the pedals 32 relative to the respective cranks 48 evolves through the kinematic trapezoid formed by the crank 48 and the lever 52. However, this evolution occurs over an angular range consistent with the permissible angles of the ankles of the pilots.

The device for displacing and locking 92 the carriage 90 to the slide 88 preferably comprises an adjustment screw 96.

The adjustment screw 96 is, for example, a worm screw.

For example, the adjustment screw 96 is able to cooperate with the carriage 90 to drive in translation the carriage 90 relative to the slide 88 along the adjustment direction when the adjustment screw 96 is rotated relative to the slide 88.

For this purpose, the carriage 90 presents a drive opening receiving the adjusting screw 96.

In particular, the adjusting screw 96 is also configured to lock the carriage 90 in position. In this case, the adjusting screw 96 is irreversible.

In particular, the irreversibility of the adjustment screw 96 is then sufficient to prevent the carriage 90 from moving relative to the slide 88, when the pilot presses on the pedals 32.

18

Thus, in this example, the irreversible adjustment screw 96 ensures both the function of displacement of the carriage 90 and of locking the position of the carriage 90. Alternatively, or additionally, the displacement and locking device 92 comprises another dedicated locking system, in particular separate from the adjustment screw 96.

The adjustment screw 96 extends according to the adjustment direction.

Here, the adjustment direction is parallel to the slide 88.

Advantageously, the displacement and locking device 92 also comprises an adjustment gear motor 98 and/or a manual adjustment member 100.

The adjustment gear motor 98 is able to be actuated by a pilot to drive in rotation the adjustment screw 96 relative to the slide 88.

The manual adjustment member 100 is able to transmit, to the adjustment screw 96, a manual torque exerted by a pilot to drive in rotation the adjustment screw 96 relative to the slide 88.

Thus, even in the event of failure of the adjustment gear motor 98, the pilot is always able to adjust the ergonomics of the rudder pedal 22 by actuating the manual adjustment member 100.

In the preferred embodiment comprising this ergonomic adjustment system 44, the mechanical kinematic chain 38 of the rudder pedal 22 is able to transmit to the output shaft 36 the displacement of the pedals 32 relative to the frame 28 for any position of the carriage 90 on the slide 88.

In the mechanical kinematic chain 38, the transmission mechanism 66 joining the central transmission part 62 to the output shaft 36 then preferably comprises a sliding connection 102 interposed between the two universal joints 74.

In the example shown in FIG. 6, the sliding connection 102 is joined to the central transmission part 62 by one of the universal joints 74 and is joined to the output shaft 36 by the other of the universal joints 74.

The sliding connection 102 presents a sliding axis.

The sliding axis passes, for example, through both universal joints 74.

The sliding connection 102 is able to be rotated as a unit, about the sliding axis, relative to the central transmission part 62 and relative to the output shaft 36.

Thus, as the carriage 90 is displaced along the direction of adjustment, the sliding connection 102 allows for a decrease or increase in the distance between the central transmission part 62 and the output shaft 36, while allowing the transmission of the rotation of the central part 62, relative to the frame 28, to the output shaft 36.

In a preferred embodiment, shown in FIG. 6, the sliding connection 102 comprises a sleeve 104 and a drive rod 106. In other words, the drive rod 106 is able to slide along the sleeve 104 according to the sliding axis.

In the illustrated example, the central transmission part 62 is joined to the sleeve 104 by one of the two universal joints 74 and the output shaft 36 is joined to the drive rod 106 by the other of the two universal joints 74.

The drive rod 106 is partially received in the sleeve 104.

In order for the sliding connection 102 to be rotated as a single unit, about the sliding axis, the sleeve 104 presents an inner surface advantageously cooperating with an outer surface of the drive rod 106 to block any rotation about the sliding axis of the drive rod 106 relative to the sleeve 104.

In the illustrated example, the sleeve 104 presents a closed cross section.

19

Preferably, the inner surface of the sleeve **104** and the outer surface of the drive rod **106** present non-circular cross-sections, in at least one region where said surfaces are in contact.

In one advantageous embodiment, illustrated in FIG. 6, the drive rod **106** and the sleeve **104** are splined.

In particular, the drive rod **106** and the sleeve **104** present corresponding splines.

The splines extend parallel to the sliding axis.

The shape of the splined drive rod **106** and the sleeve **104** allow for locking of the rotation of the drive rod **106** relative to the sleeve **104** about the sliding axis and transmission of high torques from the center piece to the output shaft **36**.

The advantageous function of braking by the rudder pedal **22** will now be described.

In this preferred embodiment, at least one of the pedal side systems **30** of the rudder pedal **22** also includes a braking system **108**.

Advantageously, each pedal side system **30** comprises a braking system **108** such as described below.

Each pedal side system **30** then carries its own braking system **108**, which allows redundancy of the braking control within the rudder pedal **22** and avoids depending on a central station common to both pedal side systems **30**, safety is improved. It is understood that it is also possible to avoid depending on a central acquisition station common to the two rudder pedals **22** for braking.

Each braking system **108** is able to control the braking of the aircraft **10**, when the wheels of the aircraft **10** are in contact with the ground.

Each braking system **108** comprises the pedal **32** of the associated pedal side system **30**, the pedal **32** then being connected to the crank **48** by a braking pivotal link **110**.

In a preferred embodiment, the braking system **108** also comprises a brake stop, the brake stop defining a stop position of the pedal **32**.

Each braking system **108** also comprises a braking acquisition system **112**.

Preferably, each braking system **108** further comprises a braking force restitution system **114**.

In the illustrated embodiment, the pedal **32** is able to be rotated relative to the crank **48** about an axis of rotation **A9** passing through said braking pivotal link **110**.

Such a rotation of the pedal **32** relative to the crank **48** about the axis of rotation **A9** corresponds, for example, to a rotation of the foot of the pilot about his heel and is intended to control a braking of the aircraft **10**. The pilot then presses a zone of the pedal **32** offset relative to the axis **A9**.

The pedal **32** is able to be rotated relative to the crank **48** about the axis of rotation **A9** independently of any joint rotation of the pedal **32** and the crank **48** relative to the frame **28** according to the axis of rotation **A1**.

In other words, the pilot can control the yaw angle and the braking independently by two distinct movements of his foot. There is no coupling of the yaw control and the braking control.

Moreover, the two pedals **32** of the rudder pedal **22** are able to be rotated relative to their respective cranks **48** about their respective axes of rotation **A9** independently of each other.

In other words, there is no coupling of the braking by the pedals **32**.

The axis of rotation **A9** of the pedal **32** relative to the crank **48** is advantageously parallel to the axis of rotation **A1** of the crank **48** relative to the frame **28**. Alternatively, the axis of rotation **A9** presents a non-zero angle, for example of

20

a few degrees, with the axis of rotation **A1**, with the pedal **32** being more oriented towards the pilot than the axis of rotation **A1**.

Preferably, the braking pivotal link **110** of the pedal **32** is concurrent with said second pivotal articulation connection **56** of the crank **48** with the support link **54**.

In other words, the axis of rotation **A2** of the crank **48** relative to the support link **54** is substantially coincident with the axis of rotation **A9** of the pedal **32** relative to the crank **48**.

The braking acquisition system **112** is configured to generate an electrical signal representative of a displacement of the pedal **32** relative to the crank **48** about the axis of rotation **A9**.

The braking acquisition system **112** is supported by at least one of the crank **48**, the lever **52**, and the support link **54**. Preferably, in the example shown in FIG. 7, the braking acquisition system **112** is supported by the support link **54**.

To accomplish this, the braking acquisition system **112** comprises at least one acquisition sensor **116**. Preferably, the braking acquisition system **112** comprises at least two redundant acquisition sensors **116**. Even more preferably, the braking acquisition system **112** comprises at least three redundant acquisition sensors **116**. In the preferred embodiment of FIGS. 7 and 8, the braking acquisition system **112** comprises at least four redundant acquisition sensors **116**.

The braking acquisition system **112** also comprises a joint drive device **118** of the acquisition sensors **116**.

Preferably, the braking acquisition system **112** comprises a system **120** for disabling the acquisition sensors **116**.

In the example shown in FIG. 7, the braking acquisition system **112** also comprises a cover, covering the acquisition sensors **116**.

Each acquisition sensor **116** comprises a fixed element and a movable element, the movable element being able to be displaced relative to the fixed element.

Each fixed element is preferably integral with one of the crank **48**, the lever **52**, and the support link **54**, for example the support link **54**.

In particular, each acquisition sensor **116** comprises a roller **122** integral with the movable element.

Each acquisition sensor **116** is able to generate an electrical measurement signal as a function of the position of the movable element relative to the fixed element over a useful electrical measurement travel.

Each redundant acquisition sensor **116** preferably presents the same useful electrical measurement travel.

For each redundant acquisition sensor **116**, an electrical measurement signal is generated in parallel. Here also, by "parallel generated measurement signals" or "parallel measurement signals" is meant signals generated by each of the redundant acquisition sensors **116** for the same relative displacement of the movable elements by the joint drive device **118**.

The acquisition sensors **116** are preferably rotatable.

Advantageously, each acquisition sensor **116** is a resistive sensor, for example a potentiometer, the fixed element comprising a resistive track and the movable element then comprising a slider. Alternatively, each acquisition sensor is an inductive sensor, for example an RVDI (Rotary Variable Differential Transformer) sensor, the fixed element then comprising at least one winding, preferably at least one primary winding and one secondary winding, the movable element then comprising a core.

The joint drive device **118** is able to simultaneously displace the movable elements relative to the respective fixed elements by the same relative displacement.

21

The joint drive device **118** is able to transform the rotation of the pedal **32** relative to the crank **48** about the axis of rotation **A9** into a joint displacement of the movable elements relative to the respective fixed elements.

To accomplish this, the joint drive device **118** preferably comprises a joint drive frame **124** for the rollers **122** of the acquisition sensors **116**.

In the example shown in FIG. 7, the joint drive device **118** further comprises an actuator arm **126** of the drive frame **124**.

The drive frame **124** is movable relative to the fixed elements of the acquisition sensors **116** and delimits, for each roller **122**, a reception housing receiving the roller **122**.

In particular, the drive frame **124** comprises a fork for each roller **122**, the fork delimiting said reception housing for the roller **122**.

The reception housing is here an open groove.

In the illustrated example, the drive frame **124** forms a cross. Any other shape could be considered, however.

In the preferred example where the acquisition sensors **116** are rotatable, the drive frame **124** is able to be rotated relative to the fixed elements of the acquisition sensors **116**, about a predetermined axis of rotation.

The predetermined axis of rotation preferably passes through a geometric center of the drive frame **124**, the geometric center then being located at the same distance from each roller **122**. Alternatively, at least two of the rollers **122** are respectively arranged at different distances from said predetermined axis of rotation of the drive frame **124**.

In the example shown in FIG. 8, the predetermined axis of rotation of the drive frame **124** relative to the fixed elements of the acquisition sensors **116** is parallel to the axis of rotation **A9** of the pedal **32** relative to the crank **48**.

For each acquisition sensor **116**, rotation of the drive frame **124**, relative to the fixed element, displaces the movable element relative to the fixed element of the acquisition sensor **116**.

More specifically, rotation of the drive frame **124** simultaneously displaces the movable elements relative to the respective fixed elements by the same relative displacement.

The actuator arm **126** is able to transform the rotation of the pedal **32** relative to the crank **48** about the axis of rotation **A9** into rotation of the drive frame **124** relative to the fixed elements of the acquisition sensors **116** about the predetermined axis of rotation of the drive frame **124**.

To accomplish this, the actuator arm **126** is connected on the one hand to the pedal **32** and is on the other hand integral with the drive frame **124**.

Preferably, the actuator arm **126** presents a point of articulation **128** with the pedal **32**.

In other words, the actuator arm **126** is able to be rotated relative to the pedal **32** about an axis of rotation **A11** passing through said point of articulation **128** of the actuator **126** with the pedal **32**.

The axis of rotation **A11** passing through said point of articulation **128** of the actuator arm **126** with the pedal **32** is preferably parallel to the axis of rotation **A9** passing through said braking pivotal link **110**.

As illustrated in FIG. 7, the actuator arm **126** is preferably angled.

In one preferred embodiment, the actuator arm **126** is further articulated.

The actuator arm **126** then comprises a first drive section **130A** connected to the pedal **32** and a second drive section **130B** integral with the drive frame **124**.

22

The first drive section **130A** is connected to the pedal **32** by means of said point of articulation **128** of the actuator arm **126** with the pedal **32**.

Preferably, the first drive section **130A** presents a pivotal articulation connection **132** with the second drive section **130B**.

In other words, the first drive section **130A** is able to be rotated relative to the second drive section **130B** about an axis of rotation **A13** passing through said pivotal articulation connection **132** of the drive sections **130A**, **130B**.

The distance between the axis of rotation **A13** of the first drive section **130A** relative to the second drive section **130B** and the axis of rotation **A11** of the actuator arm **126** relative to the pedal **32** remains constant during any displacement of the pedal **32** relative to the frame **28**. For example, the first drive section **130A** is thus rigid and non-deformable.

The second drive section **130B** is mechanically linked to the drive frame **124** and is able to rotate in conjunction with the drive frame **124**.

For example, the second drive section **130B** is rigid and non-deformable.

In one preferred embodiment, the disabling system **120** of the acquisition sensors **116** is configured, for each acquisition sensor **116**, to displace the movable element relative to the fixed element, in order to disable the useful electrical measurement travel of the acquisition sensor **116**, in the event of uncoupling of the joint drive device **118**.

In the illustrated example, the disabling system **120** exerts a disabling force on the drive frame **124**.

The disabling force is sufficient to displace, for each acquisition sensor **116**, the movable element relative to the fixed element out of the useful electrical measurement travel of the acquisition sensor **116**, in the event of uncoupling of the joint drive device **118**.

For this purpose, the disabling system **120** comprises, for example, at least one spring or spring assembly able to exert the disabling force.

In this example, the actuator arm **126** exerts, in the absence of the uncoupling of the joint drive device **118**, a holding force on the drive frame **124** opposing the disabling force. The holding force is greater than or equal to the disabling force.

In other words, in the absence of uncoupling of the joint drive device **118**, the actuator arm **126** holds each movable element within the useful electrical measurement travel of the acquisition sensor **116**. When the joint drive device **118** is uncoupled, the holding force no longer applies and the disabling force displaces each movable element accordingly.

By "uncoupling of the drive device" is meant any event from which the drive device is no longer able to jointly displace the movable elements relative to the respective fixed elements. In particular, the drive device is then no longer able to transform the rotation of the pedal **32** about the axis of rotation **A9** into a joint displacement of the movable elements relative to the respective fixed elements.

The uncoupling of the drive device refers, for example, to any failure, blockage, or breakage of a part of the joint drive device **118** or of a connection between two parts of the drive device. In particular, the term "breakage" refers to the fracturing of a solid thing into two or more parts under excessive stress or strain.

Examples include a break in the actuator arm **126**, a break in the connection of the actuator arm **126** to the pedal **32** or the drive frame **124**, or a break in a portion of the drive frame **124**.

The uncoupling of the drive device, additionally or alternatively, refers to any fault in the assembly/disassembly of

23

one or more parts of the drive device, such as, for example, the omission of a fastening screw, the loosening of one of the screws by vibration, or the misalignment of parts of the drive device.

The processing unit 26 of the control device 20 is in this example embodiment able to implement the braking function.

The processing unit 26 is connected to the braking acquisition system 112 and to at least one of the brakes of the aircraft 10.

The processing unit 26 is configured to receive each measurement signal from the brake acquisition system 112.

Specifically, in the example of the embodiment, the processing unit 26 is configured to receive the measurement signals generated in parallel by the acquisition sensors 116 of the braking acquisition system 112.

The processing unit 26 is configured to develop a brake control signal at least from each measurement signal received from the braking acquisition system 112, and in particular from the parallel measurement signals received from the braking acquisition system 112.

The processing unit 26 is then configured to send the developed braking control signal to at least one of the brakes of the aircraft 10.

In the preferred example of the present disclosure where the braking acquisition system 112 comprises the disabling system 120, of the acquisition sensor 116, the processing unit 26 is also configured to verify that each of the parallel measurement signals belongs to the useful electrical measurement travel of the acquisition sensor 116.

To accomplish this, the memory of the processing unit 26 stores, for example, characteristic information of the acquisition sensors 116, wherein the characteristic information comprises at least the useful electrical measurement travel of each acquisition sensor 116. During verification, the processing unit 26 compares each of the parallel measurement signals to the stored useful electrical measurement travel.

The processing unit 26 is configured to deduce whether the parallel measurement signals belong to the useful electrical measurement travel of the acquisition sensors 116.

Alternatively or additionally, the processing unit 26 is configured to compare measurement signals from at least two separate brake acquisition systems 112, and to deduce whether measurement signals from one of the acquisition systems 112 do not belong to the useful electrical measurement travel of the acquisition sensors 116.

The processing unit 26 is subsequently configured to generate the brake control signal from the parallel measurement signals verified as belonging to the useful electrical measurement travel of the acquisition sensors 116.

In other words, the braking control signal is therefore not generated from signals that do not belong to the useful electrical measurement travel of the acquisition sensors 116.

As a result, it is possible to rule out any measurement from a brake acquisition system 112 for which an uncoupling of the associated joint drive device 118 has occurred. This greatly improves the safety of the control device 20.

Furthermore, it is understood that the processing unit 26 is able to detect such an uncoupling of the joint drive device 118 from the described verification step.

Upon completion of the verification step, the processing unit 26 is preferably configured to inform the pilot of the detected uncoupling. For this purpose, the processing unit 26 is, for example, able to display a representative image associated with the detected uncoupling on the screen of the aircraft display system 10.

24

The braking force restitution system 114 is able to exert a force against a rotation of said pedal 32 relative to the crank 48 about the braking pivotal link 110.

The braking force restitution system 114 defines a stable position of the pedal 32 relative to the crank 48 according to the axis of rotation A9, the stable position being that adopted by the pedal 32 in the absence of external solicitation by the pilot.

The braking force restitution system 114 is supported by at least one of the crank 48, the lever 52 and the support link 54.

Advantageously, the braking force restitution system 114 and the braking acquisition system 112 are supported by the same element selected from among the crank 48, the lever 52, and the support link 54. In particular, the entire braking system 108 is carried on this same element.

Preferably, in the example shown in FIG. 7, the braking force restitution system 114 is supported by the support link 54.

In one example of the embodiment, the opposing force exerted by the yaw force restitution system 42 is preferably single slope.

In the illustrated example of the embodiment, the braking force restitution system 114 comprises, for example, at least one force member.

The force member comprises, for example, a spring. Preferably, the spring is a progressive spring.

The force member is able to exert said opposing force on the pedal 32.

Preferably, at least the force member is interposed between the pedal 32 and the support link 54 during rotation of the pedal 32 relative to the crank 48 about the braking pivotal link 110.

The pedal 32 is movable relative to the crank 48 about the braking pivotal link 110 between the stable position and the stop position against the brake stop.

The braking force restitution system 114 is separate from the braking acquisition system 112. This allows for improved safety.

In particular, the braking force restitution system 114 is arranged away from the braking acquisition system 112.

In the preferred example shown in FIG. 7, the element from among the crank 48, lever 52, and support link 54 that supports the braking system 108 (the support link 54 in this example) comprises a baseplate 134 and a wall 136 extending from the baseplate 134.

The drive frame 124 of the braking acquisition system 112 is then applied against said wall 136. The fixed elements of the acquisition sensors 116 are attached to the wall 136.

In addition, the braking force restitution system 114 presents an end that is integral with said baseplate 134.

The end of the braking force restitution system 114 is thus arranged away from the braking acquisition system 112.

In the above example, an aircraft control device 20 comprising at least two rudder pedals 22 has been described. Alternatively, the aircraft control device 20 comprises a single rudder pedal 22 as described above and is therefore devoid of a main connecting rod 24. In particular, the cockpit 16 then comprises only a single pilot seat. Indeed, many advantages related to the present disclosure remain interesting even in the case of a single rudder pedal 22.

Alternatively, to what has been described above, the central transmission part 62 is joined to the drive rod 106 by one of the universal joints 74 and the output shaft 36 is joined to the sleeve 104 by the other of the two universal joints 74.

25

As an alternative to the splined shapes of the sliding connection **102**, any shape other than splined could be suitable for blocking any rotation about the sliding axis of the drive rod **106** relative to the sleeve **104**. In another variant, one of the shaft and the sleeve **104** presents a groove that is non-perpendicular to the sliding axis and the other of the shaft and the sleeve **104** presents a key received in said groove.

Alternatively, the articulated support structure **34** of the pedal **32** comprises only the crank **48**, and is therefore devoid of the lever **52** and the support link **54** described above.

In yet another alternative, in the or each rudder pedal **22**, only one of the pedal side systems **30** of the rudder pedal **22** comprises a braking system **108** as described above.

By virtue of the mechanical kinematic chain **38** of the present disclosure, and in particular the homokinetic association of the two universal joints, the present disclosure allows the yaw angle control torque to be isolated at the end of the chain at the output shaft **36**.

This subsequently allows the two rudder pedals **22** to be connected via the main connecting rod **24**, while providing a large ergonomic adjustment range independently for each of the rudder pedals **22**.

Through the combination of the crank **48**, the lever **52** and the support link **54**, forming in particular the kinematic trapezoid, the angular orientation of each pedal **32** relative to the crank **48** about the axis **A9** is controlled, despite a large ergonomically adjustable possible travel of the operating position of the pedals **32**.

Each rudder pedal **22** can be ergonomically adapted to a wide range of pilot sizes, comprising, notably, the height from 1.57 m to 1.91 m.

The frame **28** of the rudder pedal **22** may present any possible shape.

One preferred embodiment of the frame **28** will nevertheless be described hereafter.

For the frame **28** alone, independently of the direction of the aircraft **10**, the following are defined:

- an elevation direction **Z2**, corresponding, for example, to the vertical direction **Z1** when the frame **28** is fixed in the aircraft **10**; and
- a longitudinal direction **X2** orthogonal to the elevation direction **Z2** and which is, for example, parallel to the longitudinal axis **L** of the aircraft **10**, when the frame **28** is fixed to the cockpit floor **16**;
- a lateral direction **Y2** that is orthogonal to said elevation direction **Z2** and longitudinal direction **X2**.

The frame **28** comprises a base **150**, a support framework **152**, and connection interfaces **154A**, **154B**, **154C**, each connection interface **154A-154C** being able to connect another part of the rudder pedal **22** described above to the frame **28**.

The frame **28** also comprises, for example, for each other part of the rudder pedal **22** described above, a system for securing **156** the part to the connection interface **154A-154C**.

The frame **28** presents a longitudinal median plane, the longitudinal median plane being parallel to the elevation direction **Z2** of the frame **28** and the longitudinal direction **X2**.

In particular, the median plane of the frame **28** passes through the lateral middle of the frame **28**, in other words, the middle of the frame **28** along the lateral direction **Y2**.

Preferably, the base **150**, the connecting interfaces **154A-154C**, and the support framework **152** are integrally formed as one piece, in other words, as one piece.

26

In a preferred embodiment, the base **150** assembly, the connection interfaces **154A-154C**, and the support framework **152** are integrally formed from a predetermined material and are preferably formed by a layering of said predetermined material, the predetermined material preferably being aluminum or an aluminum alloy.

The base **150** assembly, the connection interfaces **154A-154C**, and the support framework **152** are then preferably manufactured by additive manufacturing.

The base **150** is able to secure the frame **28** in the aircraft **10**.

In particular, the base **150** is able to be fixed to an inner surface of the aircraft **10**.

The inner surface of the aircraft **10** corresponds in this example to a floor of the aircraft **10**, in particular the floor of the cockpit **16** of the aircraft **10**. Alternatively, the inner surface of the aircraft **10** is separate from the floor, in which case the rudder pedal **22** is suspended, for example.

For this purpose, the base **150** delimits an outer mounting surface **158** of the frame **28** and reception holes **160** for fixing elements of the base **150**.

When the frame **28** is fixed to the inner surface of the aircraft **10** (here the floor), the outer attachment surface **158** of the base **150** is in contact with said inner surface of the aircraft **10** and fixing elements of the base **150** are received in the reception holes **160** and are fixed to the inner surface.

Furthermore, when the frame **28** is fixed to the inner surface of the aircraft **10**, the base **150** is notably interposed between the support framework **152** and said inner surface of the aircraft **10**.

The outer mounting surface **158** is, for example, flat.

The outer mounting surface **158** is in this example parallel to the longitudinal direction **X2**.

The elevation direction **Z2** of the frame **28** is in this example perpendicular to the outer mounting surface **158**.

The reception holes **160** for the fixing elements of the base **150** are through and open onto the outer mounting surface **158**.

In one preferred embodiment, the base **150** comprises at least two half bases **162**. Alternatively, the base **150** is plain.

The two half bases **162** are then separate and distant from each other.

The two half bases **162** are also arranged on either side of the longitudinal median plane of the frame **28**.

The two half bases **162** are preferably symmetrical the one from the other relative to the longitudinal median plane of the frame **28**.

The two half bases **162** extend parallel the one from the other.

In particular, each half base **162** extends along the longitudinal direction **X2**.

The support framework **152** connects each connection interface **154A-154C** to the base **150**.

The support framework **152** thus supports each connection interface **154A-154C**.

In particular, the support framework **152** of the frame **28** is able to support at least each pedal side system **30**, the mechanical kinematic chain **38**, the ergonomic adjustment system **44**, the yaw force restitution system **42**, and the yaw acquisition system **40**, when these elements are present in the rudder pedal **22** and connected by means of the respective connection interfaces **154A-154C**.

The support framework **152** is able to receive the respective load of each of these parts of the rudder pedal **22** and transfer that load to the base **150**. In other words, the support framework **152** is able to support the respective weight of each of these parts of the rudder pedal **22**.

The support framework **152** is also able to resist the various actions of the pilot on the rudder pedal **22** and in particular on the pedals **32**.

In the embodiment shown in FIGS. **9** and **10**, the support framework **152** comprises the framework elements **164** arranged in a lattice **166**, with at least one of the framework elements **164** extending from each connection interface **154A-154C**.

The lattice shape **166** of the support framework **152** provides strength optimization for maximum permissible mass and maximum permissible footprint.

The lattice **166** extends from the base **150** according to the elevation direction **Z2**.

In particular, at least two of the framework elements **164** thus extend from the base **150**, and, in the case where the base **150** comprises the two half bases **162**, at least two of the framework elements **164** extend from each half base **162**.

The lattice **166** also comprises the nodes **168** at which respectively at least two of the framework elements **164** intersect.

The lattice **166** is slender in the elevation direction **Z2** to accommodate the mechanical kinematic chain along the rudder pedal **22**.

Each framework element **164** extends along a guide.

Each framework element **164** extends along the guide between two ends.

The guide of each framework element **164** is, for example, straight or curved.

In one embodiment of the present disclosure, each framework element **164** extends along the guide from one of the nodes **168**, one of the connection interfaces **154A-154C** or the base **150**, to another of the nodes **168**, another of the connection interfaces **154A-154C** or the base **150**.

As illustrated, at least 50% of the framework elements **164** are elongated. Preferably, at least 75% of the framework elements **164** are elongated. Advantageously, all of the framework elements **164** are elongated.

By “elongated framework element” is meant that the framework element is longer than it is wide.

In one advantageous embodiment, at least 50%, preferably at least 75%, of the framework elements **164** respectively have a length at least twice, for example at least five times, greater than the largest transversal dimension.

The “length” here is taken according to the respective guide of the framework element **164**, and the term “transversal” is understood to be perpendicular to the guide.

By “greatest transversal dimension” is meant the greatest straight line distance joining two points on the outer contour of the cross section of the framework element **164**.

As illustrated in FIGS. **9** and **10**, the respective guides of at least three of the framework elements **164** extend out of the same plane. In other words, the lattice **166** is three-dimensional. Thus, the lattice **166** extends in all three directions, elevational **Z2**, longitudinal **X2** and lateral **Y2**.

Preferably, each framework element **164** presents a variable or constant transversal section along the guide of the framework element **164**.

Advantageously, each framework element **164** that extends from the base **150** presents a variable transversal section along the guide flaring toward the base **150**.

For example, each framework element **164** presents a hollow tubular transversal section or a C-shaped transversal section.

The framework elements **164** of the support frame **152** comprise the support framework elements **170** and the bracing framework elements **172**.

Each support framework element **170** is able to transmit to the base **150** the load from one of the connection interfaces **154A-154C**.

Each support framework element **170** has a guide presenting at least one component according to the elevation direction **Z2**.

For each connection interface **154A-154C**, at least one of the support framework elements **170** extends from the connection interface **154A-154C**. The connection interface **154A-154C** thus forms one end of each of its support framework elements **170**.

Each support framework element **170** is able to support the weight of the part connected to the frame **28** by means of the connection interface **154A-154C**.

Each support framework element **170** of the connection interface **154A-154C** is then in particular arranged between the base **150** and said connection interface **154A-154C**, projecting in the elevation direction **Z2** of the frame **28**.

The bracing framework elements **172** are able to stabilize the support framework **152** vis-a-vis the lateral and/or longitudinal effects resulting from actions on the frame **28**, in particular from the pilot during the actuation of the rudder pedal **22**.

The bracing framework elements **172** are able to locally stabilize at least a portion of the support framework elements **170**, for example relative to the buckling instability phenomena.

Each bracing framework element **172** has a guide presenting at least one component according to the longitudinal direction **X2** or the lateral direction **Y2**.

Each bracing framework element **172** thus preferably cooperates with at least one of the supporting framework elements **170** at a node arranged between the two ends of the supporting framework element **170**.

For example, the bracing framework elements **172** extend substantially laterally or substantially longitudinally.

In the lattice **166**, some of the bracing framework elements **172** advantageously form crosses.

In the lattice **166**, at least some of the framework elements **164** are both support and bracing framework element(s). In particular, these elements have a guide presenting at least one component according to the elevation direction **Z2** and at least one component according to the longitudinal direction **X2** or the lateral direction **Y2**.

In the lattice **166**, at least 50% of the nodes **168** are distributed according to a spatially symmetrical arrangement. The spatial symmetrical arrangement presents a plane of symmetry, preferably corresponding to the longitudinal median plane of the frame **28**.

Preferably, at least 75% of the nodes **168** are distributed according to the spatial symmetrical arrangement. Advantageously, all of the nodes **168** are distributed according to the spatially symmetrical arrangement.

In the lattice **166**, at least 50% of the framework elements **164** are distributed according to a spatially symmetric arrangement. The spatial symmetrical arrangement presents a plane of symmetry, preferably corresponding to the longitudinal median plane of the frame **28**.

Preferably, at least 75% of the framing elements **164** are distributed according to the spatial symmetric arrangement. Advantageously, all of the framework elements **164** are distributed according to the spatial symmetrical arrangement.

In the preferred embodiment shown in FIGS. **9** and **10**, the lattice **166** comprises two side half lattices **174**, respectively formed by a part of the framework elements **164**.

29

The side half lattices **174** are arranged on either side of the longitudinal median plane of the frame **28**.

The side half lattice **174** are laterally connected to each other by at least one of the bracing framework element **172**.

In the example shown in FIGS. **9** and **10**, the side half lattices **174** are laterally connected by bracing framework elements **172** forming a cross. The cross is preferably centered on the longitudinal median plane of the frame **28**.

Each side half lattice **174** extends from one of the half bases **162**.

At least one of the supporting framework elements **170** of each side half lattice **174** extends substantially parallel to the longitudinal median plane.

The securing systems **156** are not shown in FIGS. **9** and **10** but are visible in FIGS. **2** to **5**.

For each part connected to the frame **28** by means of one of the connection interfaces **154A-154C**, the associated securing system **156** comprises at least one fixing member, preferably fixing members.

The part of the rudder pedal **22** so connected belongs to one of the pedal side systems **30**, the mechanical kinematic chain **38**, the ergonomic adjustment system **44**, the yaw force restitution system **42**, or the yaw acquisition system **40**.

The associated securing system **156** optionally also comprises an intermediate fixing member **176** separate from the part and separate from the connection interface **154A-154C**, the intermediate connection member **176** being interposed between the part and the connection interface **154A-154C**.

When the part is connected to the frame **28**, the connection interface **154A-154C** is in contact with the intermediate fixing member **176** and the intermediate fixing member **176** is in contact with the part.

Such intermediate fixing members **176** are used, for example, to connect the crank **48** and the lever **52** to the corresponding connection interfaces **154A-154C**, respectively. Each intermediate fixing member **176** then defines one of the corresponding pivotal articulation connections, with the associated axis of rotation passing through the intermediate fixing member **176**.

In the event that the securing system **156** is devoid of such an intermediate fixing member **176**, the connection interface **154A-154C** is in contact with the part, when this is connected to the frame **28**.

As noted above, each connection interface **154A-154C** is able to connect to another part of the rudder pedal **22**.

In other words, the load from said other part is transmitted to the base **150** by means of the associated connection interface **154A-154C**.

Generally speaking, each connection interface **154A-154C** is either male or female.

The respective male or female nature of each connection interface **154A-154C** is independent of the respective male or female nature of each other interface. In other words, all of the interfaces may be female (as illustrated in FIGS. **9** and **10**), all of the interfaces may be male, or any non-zero number of the interfaces may be female and the remainder of the interfaces being male.

Thus, each connecting interface **154A-154C** is able to be inserted into a distinct element (male character) or to receive the distinct element (female character) to connect the part of the rudder pedal **22** associated with the frame **28**.

More specifically, said distinct element is for example the part itself, where the part is directly fixed to the connection interface **154A-154C** and therefore in contact with the connection interface **154A-154C**. Alternatively, said element is, for example, the intermediate fixing member **176** of

30

the securing system **156**, when the part is connected to the frame **28** by means of the intermediate fixing member **176** and the connection interface **154A-154C**.

Each male connection interface **154A-154C** comprises a projecting portion able to be inserted into said element to connect the part of the rudder pedal **22** associated with the frame **28**.

The projecting portion is, for example, cylindrical.

Each female connection interface **154A-154C** delimits a reception housing **178** able to receive said element for connecting the part of the rudder pedal **22** associated with the frame **28**.

The reception housing **178** is in the form of a cross.

The reception housing **178** preferably corresponds to a bore. In particular, the reception housing **178** is continuous.

For example, the reception housing **178** is cylindrical.

In addition, each connection interface **154A-154C** comprises at least one fixing region **180**.

The fixing region **180** delimits at least one reception hole for each member of the associated securing system **156**. As illustrated in the figures, the fixing region **180** preferably delimits at least two reception holes for a fixing member, advantageously at least three reception holes, more preferably at least four reception holes.

When the part of the rudder pedal **22** is connected to the frame **28** by means of the associated connection interface **154A-154C**, the fixing region **180** is in contact with said part or with the intermediate fixing member **176** and the fixing members are received in the reception holes while also being fixed to the part or intermediate fixing member **176**.

The fixing region **180** surrounds the reception housing **178** or the projecting portion of the connection interface **154A-154C**.

The fixing region **180** is preferably planar.

The fixing region **180** presents an outer contour and an inner contour.

The reception housing **178** or projecting portion extends from the inner contour of the fixing region **180**.

Thus, by the phrase "a framework element extends from the connection interface" is meant that the framework element **164** extends from the fixing region **180**, and in particular from the outer contour of the fixing region **180**.

Examples of connection interfaces **154A-154C** for parts of the rudder pedal **22** will now be described.

In one preferred embodiment, the connection interfaces **154A-154C** comprise at least two pedal interfaces **154A**, each pedal interface **154A** corresponding to a connection interface of one of the pedal articulated support structures **34** of the rudder pedal **22**. The two pedal interfaces **154A** are then disposed on either side of the longitudinal median plane of the frame **28**.

These two pedal interfaces **154A** are intended to connect respectively the cranks **48** of the two pedal articulated support structures **34** of the rudder pedal **22**.

In the preferred embodiment illustrated in the figures, the connection interfaces **154A-154C** comprise at least four pedal interfaces **154A** divided into two pairs of pedal interfaces **154A**.

The spatial distribution of the four pedal interfaces **154A** is preferably symmetrical relative to said longitudinal median plane.

These four pedal interfaces **154A** are intended to connect respectively the cranks **48** and the levers **52** of the two pedal articulated support structures **34** of the rudder pedal **22**.

Each pair of pedal interfaces **154A** is able to be connected to the same pedal articulated support structure **34** of the rudder pedal **22**.

31

In other words, each pair of pedal interfaces **154A** together support the weight of the same pedal articulated support structure **34** of the rudder pedal **22** and the pedal associated with that articulated structure.

For each pair, the pedal interfaces **154A** of the pair are arranged on the same side of the longitudinal median plane of the frame **28**.

For each pedal interface **154A**, the reception housing **178** (in the case of a female pedal interface) or the projecting portion (in the case of a male pedal interface) is advantageously rotationally cylindrical.

The inner contour of the fixing region **180** of each pedal interface **154A** is circular.

Additionally, the outer contour of the fixing region **180** of each pedal interface **154A** is preferably circular.

One of the pedal interfaces **154A** of the pair forms a distal pedal interface **154A1** and the other forms a proximal pedal interface **154A2**.

When projected in the longitudinal direction **X2**, the distal pedal interface **154A1** is arranged further from the seat of the pilot than the proximal pedal interface **154A2**.

In the pair of pedal interfaces **154A**, the distal pedal interface **154A1** is able to be connected to the lever **52** of the associated pedal articulated support structure **34** and the proximal pedal interface **154A2** of the pair is able to be connected to the crank **48** of the associated pedal articulated support structure **34**.

When the pedal articulated support structure **34** is connected to the frame **28**, the axis of rotation **A3** of the lever **52** relative to the frame **28** thus passes through the distal pedal interface **154A1**.

When the pedal articulated support structure **34** is connected to the frame **28**, the axis of rotation **A1** of the crank **48** relative to the frame **28** thus passes through the proximal pedal interface **154A2**.

The support framework **152** is able to resist the actions of the pilot on the pedals **32**. In particular, the support framework **152** is able to resist a force of at least 100 daN, preferably at least 130 daN, exerted by means of the distal and proximal pedal interfaces.

In particular, the distal pedal interface **154A1** is connected to the base **150** by two support framework elements **170** able to transmit the load carried by the distal pedal interface **154A1** to the base **150**, the two support framework elements **170** diverging from the distal pedal interface **154A1**. Each of these two support framework elements **170** extends from the distal pedal interface **154A1** to the base **150**.

In particular, the distal pedal interface **154A1** is so connected to one of the pedal half bases **162**.

When projected in the longitudinal direction **X2**, the distal pedal interface **154A1** overlaps with the base **150**. In the example of FIGS. 9 and 10, when projected in the longitudinal direction **X2**, the distal pedal interface **154A1** is arranged in the longitudinal middle of the base **150**.

The proximal pedal interface **154A2** is longitudinally arranged beyond the base **150**.

The proximal pedal interface **154A2** thus protrudes out of plumb from the base **150**.

In other words, when projected in the longitudinal direction **X2**, the proximal pedal interface **154A2** does not overlap the base **150**.

Furthermore, when projected in the longitudinal direction **X2**, the proximal pedal interface **154A2** is arranged away from the base **150**.

The proximal pedal interface **154A2** is arranged above the distal pedal interface **154A1**.

32

In other words, when projected in the elevation direction **Z2** of the frame **28**, the distal pedal interface **154A1** is arranged between the proximal pedal interface **154A2** and the base **150**.

In the pair of interfaces, the diameter of the proximal reception housing **178** (in the case of a proximal female pedal interface) or the proximal protrusion (in the case of a proximal male pedal interface) is, for example, greater than the diameter of the distal reception housing **178** (in the case of a distal female pedal interface) or the distal protrusion (in the case of a distal male pedal interface).

Preferably, the width between the two proximal pedal interfaces **154A2** is less than the width between the two distal pedal interfaces **154A1**. Here, for example, it is a question of the width between the fixing regions of the interfaces. The width here is taken according to the lateral direction **Y2**.

Furthermore, as illustrated in FIGS. 9 and 10, the two proximal pedal interfaces **154A2** are advantageously connected laterally by bracing framework elements **172** forming a cross. The cross is arranged between the two proximal pedal interfaces **154A2**.

The cross is preferably centered on the longitudinal median plane of the frame **28**.

In one preferred embodiment, the connection interfaces **154A-154C** comprise at least one sensor interface **154B** corresponding to a connection interface of the yaw acquisition system **40** of the rudder pedal **22**.

In the example of FIGS. 9 and 10, the reception housing **178** (in the case of a female sensor interface) or the projecting portion (in the case of a male sensor interface) defines a shoulder.

Advantageously, the reception housing **178** (in the case of a female sensor interface) or the projecting portion (in the case of a male sensor interface) is rotationally cylindrical.

In particular, the inner contour of the fixing region **180** of the sensor interface **154B** is circular.

As illustrated in FIGS. 9 and 10, for each pedal interface **154A**, one of the framework elements **164** connects the pedal interface **154A** to the sensor interface **154B** by extending from the pedal interface **154A** to the sensor interface **154B**.

The sensor interface **154B** is arranged above each distal pedal interface **154A1**.

In other words, each distal pedal interface **154A1** is arranged between the sensor interface **154B** and the base **150** in projection according to the elevation direction **Z2** of the frame **28**.

In one preferred embodiment, the connection interfaces **154A-154C** comprise at least one adjustment interface **154C** corresponding to a connection interface of the ergonomic adjustment system **44** of the rudder pedal **22**.

The adjustment interface **154C** is also able to allow the pilot access to the adjustment screw **96**.

The reception housing **178** (in the case of a female adjustment interface) or the projecting portion (in the case of a male adjustment interface) is oblong cylindrical. The corresponding shape is thus longer than it is wide and terminated by two half cylinders.

In particular, the inner contour of the fixing region **180** of the adjustment interface **154C** is oblong. The inner contour is thus longer than it is wide and presents rounded corners.

In addition, the outer contour of the fixing region **180** of the adjustment interface **154C** is preferably rectangular and has rounded corners.

For example, the adjustment interface **154C** is centered on the longitudinal median plane of the frame **28**.

33

The adjustment interface **154C** is, for example, longitudinally arranged beyond the base **150**.

Thus, the adjustment interface **154C** protrudes out of the plumb line of the base **150**.

In other words, when projected in the longitudinal direction **X2**, the adjustment interface **154C** does not overlap with the base **150**.

Furthermore, when projected in the longitudinal direction **X2**, the adjustment interface **154C** is arranged away from the base **150**.

As illustrated in FIGS. **9** and **10**, the adjustment interface **154C** is connected to the base **150** by at least two support framework elements **170** able to transmit the load carried by the adjustment interface **154C** to the base **150**.

Each of these two support framework elements **170** of the adjustment interface **154C** extends from the adjustment interface **154C** to the base **150**.

In the preferred example illustrated, at least one of the support framework elements **170** of the adjustment interface **154C** connects the adjustment interface **154C** to a first of the two half bases **162**, and at least one other of the support framework elements **170** of the adjustment interface **154C** connects the adjustment interface **154C** to the second of the two half bases **162**.

In the example of FIGS. **9** and **10**, each support framework element **170** of the adjustment interface **154C** extends from one of the rounded vertices of the outer contour of the fixing region **180** of the adjustment interface **154C**, respectively.

Said two support framework elements **170** of the adjustment interface **154C** are then advantageously laterally connected by the bracing framework elements **172** forming at least one cross, and in the illustrated example forming at least two crosses.

Each cross is preferably centered on the longitudinal median plane of the frame **28**.

In one embodiment, the adjustment interface **154C** is longitudinally arranged beyond the proximal pedal interface **154A2**.

The adjustment interface **154C** is arranged below each proximal pedal interface **154A2**.

In other words, for each proximal pedal interface **154A2**, the adjustment interface **154C** is arranged between the proximal pedal interface **154A2** and the base **150** in projection according to the elevation direction **Z2** of the frame **28**.

In the preferred embodiment, the support framework **152** also delimits a central functional passage **184**. Thus, the central functional passage **184** is delimited by the framework elements **164** of the lattice **166**.

In particular, the central functional passage **184** is laterally bounded by the two side half lattices **174**.

The central functional passage **184** extends longitudinally and in the elevation direction **Z2**.

The central functional passage **184** opens longitudinally to the adjustment interface **154C**.

The central functional passage **184** is thus able to receive at least the slide **88** and the carriage **90** of the adjustment system.

The central functional passage **184** also opens, in the elevation direction **Z2**, to the sensor interface **154B**.

The central functional passage **184** is able to receive at least a part of the mechanical kinematic chain of the rudder pedal **22**. In particular, the central functional passage **184** is, for example, at least able to receive at least the central transmission part **62** and the transmission mechanism **66** joining the central transmission part **62** to the output shaft **36**.

34

The central functional passage **184** allows for the passage of at least one cylinder with a diameter greater than or equal to **100 mm**.

The central functional passage **184** constitutes a significant space in the center of the frame **28** to accept the translation of the carriage **90** and of a part of the mechanical kinematic chain, advantageously over the authorized adjustment travel of at least **100 mm**.

The central functional passage **184** also constitutes a space able to accommodate the adjustment gear motor **98**.

In one preferred embodiment, the connection interfaces **154A-154C** also comprise at least one wiring interface, not shown, corresponding to a connection interface of a segregation box of the rudder pedal **22**.

The segregation box of the rudder pedal **22** is able to centralize and redistribute the cables to the various connectors.

A manufacturing method for the frame **28** of the rudder pedal described above will now be described. Any manufacturing method can be considered by the skilled person.

In one preferred embodiment, the manufacturing method comprises a step of additively manufacturing the entire base **150**, the connector interfaces **154A-154C**, and the support framework **152**.

Additionally, the method advantageously comprises a step of applying a coating to the frame **28**. This step is, for example, a protection by anaphoresis.

In the preferred additive manufacturing embodiment, the entire base **150**, connection interfaces **154A-154C**, and support framework **152** are formed by a superposition of successive layers, with the layers being deposited on top of each other.

Each layer includes at least one solid area. Eventually, each layer includes void areas delimited by adjacent solid areas, depending on the shape of the base **150**, the connecting interfaces **154A-154C**, and the support framework **152**.

Additive manufacturing is particularly well suited to the manufacture of the frame **28** given the particular shape of the support framework **152**. This shape presents angles or notches that would be difficult to obtain by molding (constrained by the shape of the mold) or milling (no space for the machine head).

Additive manufacturing makes it possible to ensure stiffness and mass constraints by manufacturing in one piece, despite a slender shape, thin cross-sections and tight tolerances.

In one embodiment, the additive manufacturing step comprises at least the following steps:

- a) forming a powder layer;
- b) selectively melting at least one region of the powder layer; and
- c) repeating steps a) and b) to form the entire base **150**, connecting interfaces **154A-154C**, and supporting framework **152**.

The additive manufacturing step is, for example, implemented by a manufacturing apparatus including at least one support surface, a device for forming successive layers of powder on the support surface, a device for selectively melting the powder, and a system for moving the selective melting device relative to the support surface.

The manufacturing apparatus also includes a control unit for the forming device, the selective melting device, and the displacement system.

In step a), each powder layer formed is flat.

Each powder layer is free of macroscopic relief with a height greater than 4 times the average thickness of the layer.

35

The powder is, for example, a metallic powder. The metal powder is preferably an aluminum or aluminum alloy powder.

In particular, step a) is implemented by the forming device and the control unit controlling the forming device.

During step b), each solid area and each possible empty area of the layer is/are formed.

During step b), the metal powder is brought to its melting point.

In particular, step b) is implemented by the selective melting device, the displacement system and the control unit controlling the selective melting device and the displacement system.

In one advantageous embodiment, the additive manufacturing is by selective laser beam melting, LBM (from the English acronym "Laser Beam Melting").

The selective melting device therefore comprises a laser capable of melting and fusing the metal powder of the layer formed in the previous step a) with the previous layers.

During step b), the control unit calculates, for example from a digital model of the frame **28**, the spatial arrangement of each area to be irradiated within the powder layer formed in the previous step a) to form the base **150**, the connecting interfaces **154A-154C** and the supporting framework **152**.

Based on this, the displacement system is controlled by the control unit to position the laser of the selective melting device relative to the layer formed in the previous step a) to irradiate each area to be formed of the layer.

A method of assembling a rudder pedal **22** will now be described.

The method comprises providing a frame **28** as described above. The frame **28** is advantageously obtained from the manufacturing method described above.

The method further comprises providing at least one part of the rudder pedal **22**.

The method then comprises, for each part of the rudder pedal **22**, connecting the part to at least one of the connection interfaces **154A-154C** of the frame **28**.

As a result of the previously described features, the frame **28** meets the requirements defined above for the frame **28** alone. In particular, the frame **28** is sufficiently compact to meet the allocated footprint, and light enough to not exceed the mass allocation of the rudder pedal **22**.

In addition, the frame **28** thus allows fixing to the floor of the aircraft **10** to improve mounting conditions and thus safety.

In the foregoing, it is clear to those skilled in the art that, for each pedal **32**, the pedal **32** is connected to the support rod **54** by the braking pivotal link **110**; and that the pedal **32** is able to be rotated relative to the support link **54** around the axis of rotation **A9** passing through said braking pivot connection **110**.

Moreover, in all the foregoing, for braking, when it is a question of a displacement of the pedal **32** with respect to the crank **48** around the axis of rotation **A9**, it is understood that it is also a question of a movement of the pedal **32** relative to the support link **54** around the axis of rotation **A9**.

In particular, the crank **48** and the support link **54** are able to be stationary with respect to each other, during the rotation of the pedal **32** with respect to the support link **54** and with respect to the crank **48** around the axis of rotation **A9** passing through said braking pivotal link **110**.

For example, the braking acquisition system **112** is configured to generate an electrical signal representative of a movement of the pedal **32** relative to the support link **54** around the axis of rotation **A9**.

36

What is claimed is:

1. An aircraft rudder pedal comprising:
a frame; and

at least one pedal side system, the or each pedal side system comprising a pedal and an articulated support structure of the pedal connecting the pedal to the frame; the articulated support structure of said pedal comprises a crank, a lever and a support link,

the crank presenting a first pivotal articulation connection to the frame and a second pivotal articulation connection to the support link, the lever presenting a first pivotal articulation connection to the frame and a second pivotal articulation connection to the support link,

wherein the lever rotatable relative to the frame about an axis of rotation passing through the first pivotal articulation connection of the lever with the frame, and the lever rotatable relative to the support link about an axis of rotation passing through the second pivotal articulation connection of the lever with the support link,

wherein the distance between the axis of rotation of the lever relative to the frame and the axis of rotation of the lever relative to the support link remains constant during any displacement of the pedal relative to the frame.

2. The rudder pedal according to claim 1, wherein the crank and the lever form a kinematic trapezoid, a line passing through the first pivotal articulation connection of the crank with the frame and through the second pivotal articulation connection of the crank with the support link being substantially parallel to a line passing through the first pivotal articulation connection of the lever with the frame and through the second pivotal articulation connection of the lever with the support link.

3. The rudder pedal according to claim 2, wherein a distance between an axis of rotation of the crank relative to the frame and an axis of rotation of the crank relative to the support link is greater than a distance between an axis of rotation of the lever relative to the frame and an axis of rotation of the lever relative to the support link.

4. The rudder pedal according to claim 1, wherein the at least one pedal side system is two pedal side systems, the articulated support structures of the two pedal side systems being arranged on either side of the frame.

5. The rudder pedal according to claim 1, wherein the or each pedal side system comprises a braking system configured to control braking of the aircraft, the braking system comprising the pedal of the pedal side system, said pedal being connected to the crank by a braking pivotal link.

6. The rudder pedal according to claim 5, wherein the braking system comprises a braking acquisition system configured to generate an electrical signal representative of a displacement of said pedal relative to the crank about the braking pivotal link, the braking acquisition system being supported by at least one of the crank, the lever and the support link.

7. The rudder pedal according to claim 6, wherein the braking acquisition system is supported by the support link.

8. The rudder pedal according to claim 6, wherein the braking acquisition system comprises at least two redundant acquisition sensors, each acquisition sensor comprising a fixed element and a movable element, the movable element being configured to be displaced relative to the fixed element, each acquisition sensor being configured to generate an electrical measurement signal as a function of a position of the movable element relative to the fixed element; the braking acquisition system also comprising a joint drive device for the acquisition sensors, the joint drive device

37

being configured to displace, for each acquisition sensor, the movable element relative to the fixed element of the acquisition sensor.

9. The rudder pedal according to claim 8, wherein the joint drive device is configured to transform a displacement of the pedal relative to the crank about the braking pivotal link into a joint displacement of the movable elements relative to the respective fixed elements.

10. The rudder pedal according to claim 8, wherein each acquisition sensor comprises a roller integral with the movable element, the joint drive device comprising a joint drive frame of the rollers of the acquisition sensors, the joint drive frame being movable relative to the fixed elements of the acquisition sensors and delimiting, for each roller, a reception housing receiving said roller.

11. The rudder pedal according to claim 10, wherein the reception housing is a groove and/or wherein the acquisition sensors are rotatable and the joint drive frame is configured to be displaced in rotation relative to the fixed elements of the acquisition sensors.

12. The rudder pedal according to claim 10, wherein the joint drive device further comprises an actuator arm of the joint drive frame, the actuator arm being configured to transform a rotation of the pedal relative to the crank into rotation of the joint drive frame relative to the fixed elements of the acquisition sensors, the actuator arm being connected to the pedal and integral with the joint drive frame.

13. The rudder pedal according to claim 12, wherein the actuator arm is articulated and comprises a first drive section connected to the pedal and a second drive section integral with the joint drive frame, the first drive section presenting a pivotal articulation connection with the second drive section.

14. The rudder pedal according to claim 8, wherein each acquisition sensor is configured to generate an electrical measurement signal as a function of the position of the movable element relative to the fixed element over a useful electrical measurement travel, the braking acquisition system further comprising a disabling system configured, for each acquisition sensor, to disable the movable element relative to the fixed element out of the useful electrical measurement travel of the acquisition sensor, in an event of uncoupling of the drive device.

15. The rudder pedal according to claim 14, wherein each acquisition sensor comprises a roller integral with the movable element, the joint drive device comprising a joint drive frame of the rollers of the acquisition sensors, the joint drive frame being movable relative to the fixed elements of the acquisition sensors and delimiting, for each roller, a reception housing receiving said roller;

and, wherein the disabling system exerts a disabling force on the joint drive frame, the disabling force being sufficient to displace, for each acquisition sensor, the movable element relative to the fixed element out of the useful electrical measurement travel of the acquisition sensor, in the event of uncoupling of the drive device.

16. The rudder pedal according to claim 5, wherein the braking system also comprises a braking force restitution system configured to exert a force against a rotation of said pedal relative to the crank about the pivotal braking link, the braking force restitution system being supported by at least one of the crank, the lever and the support link, preferably by the support link.

17. The rudder pedal according to claim 16, wherein the braking force restitution system is supported by the support link.

38

18. The rudder pedal according to claim 16, wherein the braking system comprises a braking acquisition system configured to generate an electrical signal representative of a displacement of said pedal relative to the crank about the braking pivotal link, the braking acquisition system being supported by at least one of the crank, the lever and the support link,

wherein the braking acquisition system comprises at least two redundant acquisition sensors, each acquisition sensor comprising a fixed element and a movable element, the movable element being configured to be displaced relative to the fixed element, each acquisition sensor being configured to generate an electrical measurement signal as a function of a position of the movable element relative to the fixed element; the braking acquisition system also comprising a joint drive device for the acquisition sensors, the joint drive device being configured to displace, for each acquisition sensor, the movable element relative to the fixed element of the acquisition sensor,

wherein each acquisition sensor comprises a roller integral with the movable element, the joint drive device comprising a joint drive frame of the rollers of the acquisition sensors, the joint drive frame being movable relative to the fixed elements of the acquisition sensors and delimiting, for each roller, a reception housing receiving said roller,

and, wherein one of the crank, the lever, and the support link comprises a baseplate and a wall projecting from the baseplate, the joint drive frame of the braking acquisition system being mounted against the wall, the braking force restitution system presenting an end integral with the baseplate.

19. An aircraft rudder pedal comprising:

a frame; and
at least one pedal side system, the or each pedal side system comprising a pedal and an articulated support structure of the pedal connecting the pedal to the frame; the articulated support structure of said pedal comprises a crank, a lever and a support link,

the crank presenting a first pivotal articulation connection to the frame and a second pivotal articulation connection to the support link, the lever presenting a first pivotal articulation connection to the frame and a second pivotal articulation connection to the support link,

wherein the crank and the lever form a kinematic trapezoid, a line passing through the first pivotal articulation connection of the crank with the frame and through the second pivotal articulation connection of the crank with the support link being substantially parallel to a line passing through the first pivotal articulation connection of the lever with the frame and through the second pivotal articulation connection of the lever with the support link;

wherein, during any displacement of the pedal relative to the frame, a distance between an axis of rotation of the crank relative to the frame and an axis of rotation of the crank relative to the support link is greater than a distance between an axis of rotation of the lever relative to the frame and an axis of rotation of the lever relative to the support link.

20. An aircraft rudder pedal comprising:

a frame; and
at least one pedal side system, the or each pedal side system comprising a pedal and an articulated support structure of the pedal connecting the pedal to the frame;

39

the articulated support structure of said pedal comprises a crank, a lever and a support link,

the crank presenting a first pivotal articulation connection to the frame and a second pivotal articulation connection to the support link, the lever presenting a first pivotal articulation connection to the frame and a second pivotal articulation connection to the support link,

wherein the at least one pedal side system comprises a braking system configured to control braking of the aircraft, the braking system comprising the pedal of the pedal side system, said pedal being connected to the crank by a braking pivotal link;

wherein the braking system comprises a braking acquisition system configured to generate an electrical signal representative of a displacement of said pedal relative to the crank about the braking pivotal link, the braking acquisition system being supported by at least one of the crank, the lever and the support link;

40

wherein the braking acquisition system comprises at least two redundant acquisition sensors, the at least two redundant acquisition sensors being supported by the same articulated support structure,

wherein each acquisition sensor comprises a fixed element and a movable element, the movable element being configured to be displaced relative to the fixed element, each acquisition sensor being configured to generate an electrical measurement signal as a function of a position of the movable element relative to the fixed element; the braking acquisition system also comprising a joint drive device for the acquisition sensors, the joint drive device being configured to displace, for each acquisition sensor, the movable element relative to the fixed element of the acquisition sensor.

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